

Reverse Entailment in Language

Nominal Potential, the Universal Noun Slot, and the Geometry of Discourse

By Maks Panner

PREFACE

1.

This book began from a discomfort with how different schools of linguistics draw the boundaries of lexical theory. On the one hand, grammatical categories are often treated as stable and intrinsic, nouns as natural candidates for argument positions, verbs as natural predicates, adjectives as natural property terms. On the other hand, ordinary linguistic practice routinely violates that neat division in a way that is not merely occasional but systematically context-driven: the same material can behave as an action, a property, a relation, or an entity depending on the construction that hosts it and the discourse task it is made to perform. In that respect, categorization in use is fluid and context-sensitive, resembling what quantum-inspired accounts emphasize: a unit's interpretive status is not fixed by its bare form alone, but is shaped by the "measurement conditions" supplied by syntactic position, constructional frame, and communicative context. Speakers do not discard categories; rather, they exploit a flexible interface in which context selects and stabilizes one profile among several available ones, turning actions into "things," properties into "objects of thought," relations into entities, and even function-words into discussable items. The descriptive facts are not hard to find; what has been harder is to state the mechanism behind this context-sensitive stabilization in a way that is both theoretically coherent and empirically testable. The central question that drove this project is therefore simple to state: how can languages and speakers so reliably treat almost any linguistic material as an entity, something that can occupy noun-slot positions, without collapsing the category system into triviality? I came to believe that we need two ingredients to answer that question. First, we need a clear account of what remains stable across uses (across contexts, media, and forms). Second, we need a disciplined diagnostic that allows us to infer category potential from real linguistic behavior rather than from stipulation.

2.

What makes this more than a stylistic observation is that the same context-sensitive stabilization shows up beyond armchair description. **Neurolinguistic evidence** suggests that nominal environments recruit shared integration and unification resources even when the input begins as verb-like or property-like material: nominalized actions and event-denoting expressions can engage networks associated with both action representations and lexical-syntactic unification, indicating that "entity-profiles" are not merely rhetorical but track stable cognitive operations. In parallel, the phenomenon has immediate **computational consequences**. NLP systems that treat category as rigid frequently mis-handle clausal subjects, complex nominals, and category-shifting constructions; those errors propagate into dependency parses and semantic role labeling, precisely where argument structure must be identified reliably. A diagnostic that explicitly detects noun-slot behavior and nominalized profiles can therefore improve parsing and role assignment, and it becomes especially valuable in multilingual and low-resource settings where overt category marking is reduced or distributed across constructions rather than morphology.

The first public form of these ideas appeared as a research article, "A Universal Noun Slot in the Linguistic Mind with Computational Application" (Oct 22, 2025), where I introduced the "Reverse Entailment Diagnostic (RED)" and argued that the nominal capacity of linguistic material is not an accident of English style, but a broadly general mechanism visible across typologically different languages. That article did what

articles are forced to do: compress. It presented the core claims, demonstrated them with selected examples, and sketched a computational direction, but it could not give sustained attention to the deeper architecture needed to make the diagnostic conceptually stable. The present book is an attempt to unfold what the article necessarily folded: to slow down and clarify the objects of analysis; to separate form, lexical organization, and citation practices; to broaden the empirical base; and to develop the cognitive and computational consequences in a way that does not read as an afterthought. In several places, the book also revises the article's presentation, - not by abandoning the central insight, but by tightening definitions, improving the inferential logic, and making explicit which conclusions follow from which premises.

3.

As the project developed, it became clear that RED is only as strong as the structural framework that hosts it. A diagnostic cannot float in isolation; it presupposes a theory of what kind of object is being diagnosed, what counts as evidence, and what kind of persistence the diagnostic is tracking across contexts. The article introduced RED as a bidirectional inference tool precisely to confirm nominal potential: the capacity of linguistic material to be nominalized, to behave as a syntactical noun entity, and to occupy noun-slot positions in well-formed structures. But those recurring “road signals” point to something deeper than a diagnostic: they presuppose a background space in which such category shifts and stabilizations are even possible, and in which “entity-like” construal is not an accident but a structured possibility of linguistic cognition.

For this reason, the book introduces the idea of a “Universal Cognitive Domain (UCD)”, also called the “Universal Linguistic Domain (ULD)”, as the “foundational geometry of linguistic activity”: an area of possible linguistic discourse, a formal space of admissible syntactic - cognitive positions within which categorization, profiling, nominalization, and interpretation take place. On this view, the “universal noun slot” is not an add-on description of flexibility, but a structural feature of that domain: a privileged region of discourse-space in which diverse linguistic material can be stabilized as an entity-like “thing” when constructions demand it. RED, in turn, is best understood as an evidential instrument operating within this geometry: it detects when particular forms, constructions, and contexts force an item to enter the noun-slot region of the domain, thereby revealing the deeper constraints and affordances that organize linguistic behavior.

This domain-based perspective also makes explicit why the book must be interdisciplinary. In “cognitive science”, the ULD proposal is a claim about the architecture of mental representation: speakers possess stable, reusable slot-structures that allow category re-profiling without loss of identity, and the noun slot is a particularly general interface for “thinghood” and reference. RED is therefore not merely a descriptive test but a probe into how the mind converts actions, properties, relations, and even function-word material into entity-like objects of thought under pressure from construction and discourse. In “computational linguistics”, the same geometry becomes operational: if linguistic discourse is structured by recurring slot-types, then systems that treat category as rigid will systematically fail at the boundaries where re-profiling occurs. RED can be implemented as a detection-and-relabeling layer that identifies noun-slot behavior and nominalized profiles, improving dependency parsing, semantic role labeling, and multilingual disambiguation—especially in languages where category marking is distributed across constructions rather than overt morphology.

Within this architecture, the book distinguishes (i) the formal layer of expression patterns, (ii) language-specific lexical organization, and (iii) the indexing conventions that lexicography and annotation impose on real data, while treating all three as operating inside the same underlying universal domain that makes linguistic activity possible in the first place. The formal layer anchors stability across modalities and realizations; lexical organization explains how languages regiment category potentials; and indexing conventions explain how real-world corpora and reference systems package lexical items for retrieval and comparison. The ULD supplies the background “space of possibilities” in which these layers interact.

A second reason for writing a book rather than a second article is audience. The universal noun slot question touches multiple communities—formal semantics, morphosyntax, cognitive linguistics, neurolinguistics, and NLP—and each of those communities expects a different kind of argument and a different standard of

explanation. A monograph allows the same core hypothesis to be presented through multiple lenses without forcing any one community's preferred format to dominate.

4.

Readers primarily can be interested in the “big idea”: how noun-slot behavior reveals a universal mechanism. In this case, they may read the Introduction, then move directly to the chapters that introduce RED and the cross-linguistic case studies, returning to the formal architecture as needed. Readers interested in “formal structure and terminology” should read the early chapters in order, since later arguments rely on the some persistence principles. Readers interested in “NLP applications” can move quickly to the computational chapters, using the earlier material as definitions and justification for the operational criteria. Readers interested in “cognitive and neurological grounding” may focus on the chapters that connect noun-slot profiling and nominalization to broader models of categorization, representation, and unification.

More importantly, a book provides the space to introduce what the article could only gesture toward: a Universal Linguistic Domain as the area of possible discourse and a structured space of admissible syntactic–cognitive positions in which categorization and nominalization occur. Once that domain is stated explicitly, the “universal noun slot” becomes a principled claim about domain geometry rather than a loose metaphor, and RED becomes the disciplined method for detecting when items cross into that entity-region and what constraints govern their stabilization there. The same explicit domain model also makes it possible to formulate testable cognitive predictions and implementable computational criteria, rather than leaving the proposal at the level of interpretive rhetoric.

A book-length treatment also provides room to address the predictable objections carefully: that nominalization is merely a quirk of English; that many examples are “just metalinguistic mention”; that the phenomenon is reducible to a small set of constructional types; or that it becomes trivial once one permits coercion. Some of these objections are correct in part, but only once the relevant distinctions are made—between ordinary discourse and explicit mention, between language-specific mechanisms and domain-level constraints, between unconstrained creativity and diagnostics that impose principled tests, and between descriptive possibility and cognitively licensed stabilization. The value of the book format is that it allows those distinctions to be stated explicitly, defended, and then used to separate what is merely rhetorical from what is structurally informative. It also allows neurological and computational evidence to be treated not as ornaments but as converging constraints: cognitive and neural results shape what the ULD can plausibly be, while computational performance reveals where rigid category assumptions break down in practice.

Finally, there was a practical impetus. When a press expressed interest in seeing this project developed into a full-length monograph, it provided the external structure necessary to treat the work as a book rather than as an indefinitely expanding manuscript. That invitation helped clarify what needed to be foundational—particularly the Universal Linguistic Domain as the background space of possible discourse, the diagnostics that reveal its slot-structure, and the conceptual separation of formal, lexical, and indexing layers—what needed to be demonstrated through cross-linguistic data, cognitive and neurological convergence, and computational performance, and what could responsibly be left as open research for later refinement and broader testing.

5.

This book advances a specific framework for thinking about lexical identity and category potential, and it develops “RED” as a disciplined method for diagnosing entity-like behavior across parts of speech. It does so within a stronger organizing commitment: linguistic activity is taken to unfold inside a “Universal Linguistic Domain”, an area of possible linguistic discourse, conceived as a structured space of admissible syntactic–cognitive positions within which elements can be stabilized as entities, predicates, or modifiers depending on constructional and contextual demands. The universal noun slot hypothesis is treated as a claim about the geometry of that domain, and RED is presented as an evidential instrument for detecting when linguistic material is forced into the domain's entity-region.

It is not a general textbook in theoretical linguistics, nor is it a survey of all positions in the long debate over categorization, nominalization, or the semantics-syntax interface. It does not attempt to resolve every metaphysical question about meaning, nor does it pretend to offer a complete sociological account of language use. Its aim is more focused: to propose a coherent architecture in which (a) lexical identity can be discussed without collapsing into either pure context-dependence or rigid lexical essentialism, and (b) nominalization can be treated as a principled, testable capacity rather than as a collection of isolated “special constructions.” At the same time, the book is intentionally “interdisciplinary”: it connects the domain-level hypothesis to “cognitive science” (as a claim about stable mental slotting and entity-profiling mechanisms) and to “computational applications” (as an operational criterion that can improve parsing, semantic role labeling, and multilingual disambiguation by explicitly identifying noun-slot behavior and nominalized profiles in real data).

6.

Each chapter is written to be internally coherent, but the argument is cumulative: later diagnostics and applications are strongest when the foundational distinctions have been absorbed. This book introduces a small number of technical terms: most notably “Exeme”, “Reverse Entailment Diagnostic (RED)”, “Universal Linguistic Domain (ULD)”. These are defined explicitly in the early theoretical chapters and collected in a glossary. I have kept coinage to a minimum, but some new labels are necessary to mark distinctions that existing vocabulary often blurs, especially the distinction between modality-neutral form patterns and higher-level lexical organization.

Examples are drawn primarily from English because it offers abundant corpora and a rich descriptive tradition, but the empirical argument is explicitly cross-linguistic. When examples from other languages are used, I provide glosses and translations, and I state the relevant morphological or constructional constraints that license nominal behavior. Where judgments are involved, I separate (i) ordinary acceptability from (ii) explicitly metalinguistic or quotation-like regimes, since these behave differently and must not be conflated. Citations follow an author–date system; full bibliographic details appear in the References. Computational sections describe algorithmic ideas at a level intended to be implementable; where pseudocode is presented, it should be read as a schematic template rather than as a complete software specification.

Acknowledgements

A project that crosses theoretical linguistics, cognition, and computation inevitably accumulates debts. I am grateful to the scholars whose work made this book possible even when my conclusions diverge from theirs, because the project is written against a background formed by decades of careful debate on category, nominalization, and the architecture of the lexicon. I also thank the editorial team at the ELIVA PRESS for the encouragement to develop a dense article into a monograph and for the patience that such development requires. Finally, I thank my wife (Tetyana Morozova) without whom this project wouldn’t have been possible, the friends and readers who discussed early drafts, pressed for clarification, and insisted—sometimes bluntly—that the claims be stated in a way that a critical reader could actually test.

Chapter 1

Introduction: The Universal Noun Slot and the Geometry of Possible Discourse

As I said, this book began from a discomfort with how different schools of linguistics draw the boundaries of lexical theory, and from a tension that most theories acknowledge but rarely formalize in a single, testable architecture. On the one hand, grammatical categories are often treated as stable and intrinsic: nouns as natural

candidates for argument positions, verbs as natural predicates, adjectives as natural property terms (Chomsky, 1970). On the other hand, ordinary linguistic practice repeatedly and systematically violates that neat division: speakers routinely package actions as “things,” properties as “objects of thought,” relations as entities, and even function-words as discussable items (Lakoff & Johnson, 1980). The result is not random category chaos, but a patterned kind of contextual fluidity: the same linguistic material can be stabilized as different kinds of discourse objects depending on the construction that hosts it and the interpretive task the discourse requires (Goldberg, 1995).

This contextual fluidity has been captured in different ways in the literature. Dynamic morphosyntactic frameworks emphasize category flexibility and the role of morphosyntactic structure in category assignment; constructional approaches emphasize the licensing power of argument-structure templates; distributional and contextual models emphasize interpretive sensitivity to usage conditions (Goldberg, 1995; Halle & Marantz, 1993). Quantum-inspired accounts, in particular, provide a useful metaphor for what is happening at the level of analogy: interpretive status is not fixed by bare form alone, but is sensitive to “measurement conditions” supplied by syntactic position, constructional frame, and discourse context. In that spirit, the category a unit displays in a sentence can be treated as an outcome of stabilization under contextual constraints rather than as a property rigidly stored in the form itself (Lakoff & Johnson, 1980; Goldberg, 1995).

The descriptive facts are not hard to find; what has been harder is to state the mechanism behind this context-sensitive stabilization in a way that is both theoretically coherent and empirically testable. The problem is that, while many theories can describe this flexibility, far fewer can **diagnose** it in a disciplined way, and fewer still can connect that diagnostic to a principled account of lexical identity across contexts, media, and time. The aim of this book is to supply that missing structure.

1.1 The core phenomenon: noun-slot behavior as a structural signal

The most revealing site of category flexibility is the noun slot: subject and object positions, predicate nominals, genitive/possessive environments, and other constructions that require an entity-like discourse referent. Across languages, noun-slot environments exert a distinctive pressure: they force linguistic material, sometimes clearly verb-like, adjective-like, or function-word-like, to be construed as an entity-like “thing.” This is not merely a rhetorical trick. It is a recurrent grammatical option that speakers exploit for reference, evaluation, metalinguistic control, and conceptual reification.

The central question that drove this project is therefore simple to state: **how can languages and speakers so reliably treat almost any linguistic material as an entity—something that can occupy noun-slot positions—without collapsing the category system into triviality?** To make this claim testable rather than impressionistic, I argue that two ingredients are required. First, we need a clear account of what remains stable across uses (across contexts, media, and forms). Second, we need a disciplined diagnostic that allows us to infer category potential from real linguistic behavior rather than from stipulation..

The guiding hypothesis of this book is therefore that noun-slot behavior is not an accidental side-effect of a few idiosyncratic constructions. Rather, it is evidence of a general capacity in the linguistic mind to stabilize almost any linguistic material as an entity when grammar and discourse jointly demand such stabilization. In the terms of this project, I call that capacity **the universal noun slot**: those structural positions and environments that *require* an entity-like discourse referent, something that can be introduced, tracked, evaluated, possessed, quantified, or predicated of. In the present project, “noun slot” is therefore not a synonym for “noun” as a lexical category; it is a **constraint profile** imposed by constructions, subject and object positions, predicate nominals, genitive/possessive environments, and related configurations in which grammar and discourse jointly demand *thinghood* (in the ordinary cognitive sense of “something we can refer to”).

1.2 From “category flexibility” to a Universal Linguistic Domain

If the universal noun slot is real, it must be more than a slogan. It must be anchored in a formal conception of what kinds of discourse positions are available in the first place—what the space of possible linguistic activity looks like, and what it means for an element to “enter” an entity-profile.

For that reason, that is, to make the claim non-metaphorical, this book also introduces the **Universal Linguistic Domain (ULD)**: the *area of possible linguistic discourse*, conceived as a structured geometry of admissible syntactic–cognitive positions within which categorization, profiling, nominalization, and interpretation take place. In the ULD, categorization is not only a matter of lexical labels; it is a matter of how linguistic material is licensed to occupy particular regions of discourse-space (entity-regions, predicate-regions, modifier-regions, connective-regions, and so on) under constructional and contextual constraints. The universal noun slot is then a structural feature of this geometry: a privileged region of possible discourse (ULD) in which diverse linguistic material can be stabilized as an entity-like object or “thing” when constructional and discourse constraints demand it.

This move is crucial and has two benefits. First, it prevents “category flexibility” from collapsing into mere permissiveness. If there is an underlying domain geometry, then shifts are not arbitrary; they are constrained transitions into specific kinds of discourse positions, especially into the noun-slot region where entityhood becomes grammatically and cognitively salient. Second, it turns the universal noun slot claim into something testable: it predicts that entity-stabilization will show recurrent signatures across languages, across modalities, and across the boundary between ordinary language use and metalinguistic mention.

1.3 RED: a diagnostic for nominal potential in the universal domain

The book’s central methodological contribution is the **Reverse Entailment Diagnostic (RED)**. RED treats nominality not as a stipulative label but as an inferentially testable capacity. The idea is straightforward: if a derived form can be interpreted in a nominal environment (for example, under possessive/genitive marking or nominalizing morphology), then we can “reverse-entail” from that derived form back to the base to confirm that the base can function as an entity-like unit in a noun-slot position. In practice, this yields a disciplined procedure:

1. Apply a nominal derivation (possessive/genitive marking, nominalizing morphology, or constructional nominalization).
2. Test grammaticality and interpretability under controlled conditions (including metalinguistic regimes where appropriate).
3. Reverse-entail back to confirm noun-slot potential by placing the base in noun-slot environments.

The diagnostic is designed to do two things at once. First, it produces an empirical criterion for when category shifting is not just an impression but a grammatically licensed stabilization. Second, it functions as a probe into the ULD’s geometry: it reveals which constructions and contexts reliably force entry into the entity-region of the universal domain, and which cases are merely rhetorical, marginal, or metalinguistically exceptional.

1.4 Lexical persistence, exeme structure, and what exactly is being diagnosed

A diagnostic like RED (introduced above) cannot function without a clear ontology of the objects to which it applies. This book therefore develops an explicit three-layer architecture:

- **Exeme**: a modality-neutral expression unit—an expression pattern (e.g., r-u-n in writing, [rʌn] in speech) considered purely as a form-sequence, independent of grammatical category, lexical status, or meaning. Exemes supply the lowest formal layer for the higher lexical units.
- **Lexeme**: the language-specific lexical unit organized over one or more exemes, carrying structured semantic and distributional potential and participating in a language’s category system.
- **Lemma**: the conventional citation form used to index a lexeme in lexicography, corpora, and annotation systems.

This architecture separates three things that are routinely conflated in both theory and computational practice: (i) the form-pattern itself, (ii) the language-internal lexical organization that makes form-patterns “words” of a language, and (iii) the indexing conventions that let scholars and systems refer to lexical items consistently. When we ask whether “a verb can be a noun,” we are often mixing these layers. The ULD provides the background discourse geometry; the *exeme/lexeme/lemma* ladder provides the object-language scaffolding needed to make diagnostics precise. In fact, the argument of this book requires a stability principle—an account of what persists when linguistic material shifts profiles across constructions, contexts, and media. I call that principle **Lexical Ontology Persistence (LOP)**. In its simplest form, LOP claims that lexical identity is not rebuilt from scratch at every token of use. Rather, a lexical unit persists as an internally structured object, while context and construction select, foreground, and stabilize particular facets of that structure. This is how the same “word” can behave as an action, a property, a relation, or an entity without forcing us into either rigid lexical essentialism or pure context-dependence (Goldberg, 1995; Lakoff & Johnson, 1980).

LOP also entails a stronger claim about **plurapotentiality**. A lexical unit is not merely associated with a single categorical destiny; it carries a **universal plurapotentiality**—a structured availability of further syntactic differentiations. In other words, lexical material is equipped, in principle, to enter multiple constructional pathways (nominal, predication, modification, connective), and those pathways are not arbitrary: they are constrained by the grammar’s slot structure and by the combinatorial demands of predicate–argument organization (Dowty, 1991). This is why category-shifting phenomena recur with such regularity across languages: what varies is not the existence of the capacity, but the specific morphosyntactic resources each language uses to realize it (Halle & Marantz, 1993; Goldberg, 1995).

This is where LOP joins the other pillars of the book. The **Universal Linguistic Domain (ULD)** names the *space of admissible discourse positions*—the geometry of possible linguistic activity in which entityhood, predication, modification, and linking are available as stabilized outcomes. LOP explains what it means for something to *remain the same* while moving through that space: persistence is the identity condition; profiling is the mode of stabilization inside the domain. The **Reverse Entailment Diagnostic (RED)** is then the methodological bridge between theory and evidence. RED does not merely note that nominalization happens; it tests when an element has entered the noun-slot region of ULD by using controlled nominal environments (e.g., possessives/genitives and noun-slot occupancy) and inferring backward from the licensed derivative to the base’s nominal potential (Partee, 1997). In LOP terms, RED detects a particular stabilization event—an entity-profile—over a persisting lexical object.

Finally, the *exeme / lexeme / lemma* ladder prevents the discussion from collapsing into terminological ambiguity. The *exeme* anchors the modality-neutral form pattern; the *lexeme* is the language-specific unit whose persistent internal organization carries plurapotentiality; and the *lemma* is the indexing convention that lets reference systems point to that lexeme reliably. LOP is therefore not a claim about orthographic tokens or isolated utterances. It is a claim about the persistence of lexical organization over time and across contexts, while ULD specifies the domain of possible stabilization and RED supplies the diagnostic logic that makes “noun-slot capacity” empirically tractable (Halle & Marantz, 1993; Palmer et al., 2005).

1.5 Why this is not only theoretical: neurolinguistic convergence and parsing and semantic role labeling

If the universal noun slot is a real cognitive capacity rather than merely a descriptive convenience, we should expect it to leave traces in how the brain integrates and stabilizes linguistic material under nominal demands. This is not a claim that “the brain has a noun module.” Rather, it is the weaker and more testable expectation that **nominal environments** should recruit systematic integration pathways even when the input begins as action-like or property-like material.

Two strands of evidence motivate taking this expectation seriously. First, work on the coupling between language and action suggests that nominalized event expressions can engage both lexical–semantic and action-related networks, consistent with the idea that entity-profiling of actions has neural consequences rather than being merely rhetorical (Pulvermüller, 2005). Second, research on syntactic–semantic unification in

comprehension supports the view that nominal structures recruit stable integration mechanisms, as one would predict if noun-slot stabilization is a general interface operation rather than an idiosyncratic trick (Hagoort et al., 2004). These findings do not by themselves prove the Universal Linguistic Domain (ULD) as a part of universal mind-like sphere, but they support treating noun-slot behavior as a genuine cognitive target rather than a purely notational convenience.

The same phenomenon has immediate consequences for natural language processing. Many failures in dependency parsing and semantic role labeling occur precisely where category is not “as expected”: clausal subjects, complex nominals, nominalized phrases, and category-shifting constructions. When a system treats category as rigid, it tends to misidentify what counts as an argument, mis-assign thematic roles, and mishandle the structure of nominalized events—errors that propagate into downstream interpretation.

RED is designed to be operationalizable in this setting. It can function as a detection layer that flags noun-slot behavior and stabilizes nominalized material as entity-like units for downstream tasks. The relevance to semantic role labeling is especially direct, because role inventories (e.g., PropBank-style roles) depend on robust recovery of predicate–argument structure across surface variation (Dowty, 1991; Palmer et al., 2005). A framework that can reliably identify when a phrase behaves as an entity in a noun slot can therefore improve argument detection and role assignment, particularly in multilingual contexts where nominalization strategies vary in overt morphology but converge in slot behavior.

1.6 Why existing approaches are not enough

Several major approaches already capture parts of the phenomenon, and this book does not deny their value.

- **Lexicalist and category-rigid traditions** offer strong accounts of morphological regularities and selectional behavior, but they tend to treat noun-slot shifts as peripheral or derivative.
- **Distributed, root-based approaches** provide a powerful way to think about category as syntactically assigned, but they often do not provide a diagnostic logic that connects category shifts to a general cognitive slot structure.
- **Construction-based approaches** correctly emphasize the licensing power of templates, yet they can under-specify the deeper question of why noun-slot stabilization is so general, and why it reappears across typologically distinct languages.
- **Contextual/distributional and quantum-inspired approaches** capture interpretive fluidity extremely well, but they tend to model “collapse” or stabilization without making explicit what the underlying space of possible discourse positions must be for such stabilization to be systematic rather than ad hoc.

The present framework is designed to integrate what these approaches get right while filling what they leave implicit: the domain geometry (ULD), the object-layer architecture (exeme/lexeme/lemma), and a diagnostic methodology (RED) that converts descriptive observations into testable inferences about noun-slot capacity.

What is new here is not the observation that nominalizations exist, or that constructions can coerce category-like behavior. The novelty is the **architecture** that makes those familiar facts diagnostic. The original article introduced **RED** as a bidirectional method for confirming nominal potential in noun-slot positions; the book develops that insight into a larger system that (a) formalizes an explicit **Universal Linguistic Domain (ULD)** as the area of possible linguistic discourse—an underlying geometry of admissible syntactic–cognitive positions—(b) separates the object of analysis into **exeme / lexeme / lemma** so that form patterns, language-specific lexical organization, and citation/indexing practices are not conflated, and (c) integrates RED with both cognitive plausibility and computational implementability (Halle & Marantz, 1993; Goldberg, 1995; Partee, 1997; Palmer et al., 2005). In this way, the book treats noun-slot behavior as a **structural signal** rather than as an anecdotal curiosity: a controlled window into how linguistic material is stabilized as an entity-like discourse object.

Readers in linguistics will also see, immediately, where objections may arise—and the book takes those objections as part of the project rather than as inconveniences. Some will worry that “universal noun slot” risks rebranding well-known phenomena (nominalization, coercion, quotation/mention) without adding constraints;

others will suspect that certain edge cases depend on explicitly metalinguistic regimes and therefore do not generalize to ordinary grammar (Goldberg, 1995; Partee, 1997). Formal semanticists may question the use of “entailment” language if it is not clearly separated from propositional entailment, and syntacticians may argue that existing machinery already derives many of the relevant facts without positing a domain geometry (Chomsky, 1970; Halle & Marantz, 1993). Still others will press on evidence: whether cross-linguistic markers line up cleanly, whether acceptability relies too heavily on English-like intuitions, whether neurolinguistic findings support mechanism rather than correlation (Hagoort et al., 2004; Pulvermüller, 2005), and whether computational gains reflect a robust evaluation rather than a pilot illustration (Dowty, 1991; Palmer et al., 2005). These are fair pressures. The book’s response is methodological: it tightens the evidential standards of RED, separates ordinary-use data from mention-regime data, treats ULD as a constraint space rather than a metaphor, and makes explicit what would count as confirmation or failure for each claim.

1.7 Scope, method, and evidence

This book is primarily theoretical and conceptual, but its arguments are grounded in three classes of evidence:

1. **Cross-linguistic morphosyntax and constructional data:** a recurring set of diagnostics applied across typologically diverse languages to show that noun-slot stabilization is not an English-only artifact.
2. **Cognitive and neurological convergence:** evidence that entity-profiling and nominal stabilization recruit systematic representational and unification resources, supporting the claim that noun-slot behavior tracks a real cognitive capacity rather than a purely stylistic convention.
3. **Computational applications:** operational tests showing that noun-slot stabilization matters for parsing and semantic role labeling, and that RED-style detection can improve performance where rigid category assumptions fail.

The approach is therefore deliberately bridging: it aims to produce a framework that is linguistically explicit, cognitively interpretable, and computationally implementable.

1.8 Who this book is for and what it contributes

The primary audience consists of researchers and advanced students in theoretical linguistics, semantics, and the syntax–semantics interface, as well as scholars in cognitive science, neurolinguistics, and computational linguistics. The book also aims to be useful to practitioners whose work requires operational criteria for category behavior, including corpus linguists, lexicographers, and NLP engineers.

The contributions are intended to be concrete:

- a formal proposal for a **Universal Linguistic Domain** as the area of possible multilayered discourse,
- an explicit hypothesis of a **universal noun slot** as a structural and functional region of that domain,
- a disciplined diagnostic procedure (**RED**) for confirming nominal potential,
- a clean object-language scaffold (**exeme** / **lexeme** / **lemma**) for separating form-patterns, lexical organization, and indexing practice,
- cognitive and computational consequences that generate testable predictions and implementable criteria.

1.9 Roadmap of the book

Chapter 2 introduces the Universal Linguistic Domain as the geometry of possible discourse and motivates the universal noun slot as a domain-level feature rather than a language-specific quirk.

Chapter 3 develops Lexical Ontology Persistence as a theory of lexical identity under contextual profiling, clarifying how stability and fluidity coexist without collapsing into either pure contextualism or rigid lexical essentialism.

Chapter 4 defines *exeme*, *lexeme*, and *lemma* and shows how separating these layers prevents common confusions in both linguistic theory and NLP annotation.

Chapter 5 presents RED in full: its inferential logic, its evidential standards, and its status as a probe of the noun-slot region of the ULD.

Chapters 6 and 7 apply the diagnostic across languages and across parts of speech, including careful treatment of metalinguistic regimes and boundary cases.

Chapters 8–10 develop the cognitive science, neurolinguistic, and computational consequences, culminating in practical criteria for integrating RED into parsing and semantic role labeling pipelines, especially in multilingual and low-resource settings.

The final chapter summarizes the framework, addresses residual objections, and lays out a program of predictions and open problems.

1.10 How to read this book

Readers primarily interested in the conceptual claim about noun-slot stabilization can read Chapters 1, 2, and 5 first, then proceed to the cross-linguistic demonstrations. Readers primarily interested in cognitive grounding may focus on Chapters 2, 5, and the cognitive/neurological chapters. Readers primarily interested in computational implementation can move quickly to the parsing and SRL chapters, using earlier chapters as definitional infrastructure.

1.11 Extensions and future developments

In furtherance of the next book, I will develop the **Universal Linguistic Domain (ULD)** in detail, treating it as a **linguistic matrix**: a structured (“geometric”) space whose constraints and affordances govern what kinds of discourse-positions are available and how linguistic material can be stabilized within them. On the definition adopted here, **ULD is not a family of separate slot regions**. It is **one universal nounal slot space with internal positional geometry**. In that sense, **ULD = UNS**: ULD is the theoretical structure that explains how universal nounal hosting is organized and constrained, while **RED** is the empirical method that demonstrates that nounal hosting is universal. In operational terms, RED is the “entry test” for the noun-slot region of ULD and ULD is, therefore, the theoretical space of possible discourse; RED is the diagnostic instrument that shows when and how linguistic material moves into the noun-slot region of that space, and thereby provides empirical access to the domain’s geometry.

This matrix plays a generative role at multiple levels, and those levels are the book’s organizing order. The **most fundamental level (entry point / entry unit) is Level-0**, where **exemes** enter the empty hosting field as basic linguistic atoms. However, the **primary linguistic organization**—the minimal stabilized “molecule”—is **Level-1**, the **term (t)**: a **one-term** linguistic structure realized as a **three-region tape** (Ordinate / Subordinate / Coordinate), i.e., a three-box hosting model that yields a simple but internally articulated linguistic term.

The first genuinely relational (syntactical) organization arises when **two Level-1 terms are coupled**. The combination of two three-region tapes—equivalently, the combination of two one-term structures ($2 \times t$)—yields a **two-term structure** $t_1 R t_2$, which is the core of **Level-2** organization: a two-term / three-domain configuration (**S–P–O**) with a **special predicate connector and its axis micro-slots**. Level-2 has two structural orders that must be distinguished.

(a) Level-2a: PAS as the germinal layer (nucleating matrix). At the most fundamental syntactic level—the first-order construction in the sense relevant to derivation—this two-term configuration yields a **predicate–argument skeleton (PAS)** as the earliest syntactic derivational entity: a bare but organized family of slots, an abstract argumentative frame that structures roles, slotting, and combinatorial possibilities prior to full sentence realization (Dowty, 1991; Halle & Marantz, 1993). In this sense, PAS functions as a **generative**

matrix (a germinal / nucleating layer, a “reading frame”) produced when two one-term structures are bound through **predication**—through the predication relation RRR that converts mere hosting into a predicate–argument dependency.

(b) Level-2b: PAS-derived primary syntactic units (ready-to-go units). From PAS, the system produces modified forms of PAS—still Level-2 in kind, but higher *super-orders* of Level-2—namely the familiar primary syntactic elements that constitute the “ready-to-go” units of linguistic output: **phrases, clauses, sentences, and even certain compound words**. On this view, these units are not primitive. They are **second-order constructions**: elaborations of a first-order predicate–argument skeleton, produced by embedding, extending, and reconfiguring that skeleton into increasingly articulated structures (Goldberg, 1995). What appears as the familiar inventory of sentence types and phrasal forms is therefore understood as a family of systematic derivations from PAS rather than as a set of unrelated primitives.

Beyond Level-2, the matrix permits further composition into supra-sentential organization. The **next level of organization**—a **tertiary level** (the tertiary freedom of ULD)—is **Level-3: a four-term syntactic structure**, schematically $2 \times 2t_2 \times 2t_2 \times 2t_2$ (two two-term structures combined), formed by the combination of two Level-2 predicate–argument units. These **supra-syntactical structures (SSS)** function as organizational and functional elements of extended discourse: the first primary functional units of text built out of PAS-derived products, now capable of carrying larger scopes of coordination, modification, operator structure, and multi-clause organization.

Finally, the **Level-4** organization is **text** itself: a higher-order domain in which discourse forces—topic and thematic management, plot framing, rhetorical contrast, evaluation, and genre constraints—operate over and through syntactic structures without reducing to them. At this level, matrix “chains” (linguistic “circuits”) of derived units form a tertiary organization in which rhetorical and discourse constraints shape coherence, emphasis, and informational flow, yielding a modified, multidimensional, space-like environment of interpretation rather than a mere linear sequence of first- and second-order constructions. This discourse field interacts with the linguistic matrix by biasing which slots are foregrounded, which entity-profiles are stabilized, and how predicate–argument skeletons are chained into extended discourse.

In this way, the **Universal Linguistic Domain** is not treated as an abstract metaphor but as a **multi-level generative geometry** whose levels—Level-0 exeme entry, Level-1 term hosting (three-region tape), Level-2 PAS nucleation and PAS-derived syntactic units, Level-3 supra-syntactical four-term organization, and Level-4 text as discourse-force field—constitute the **dimensions of linguistic space**: the field of possible discourse within which linguistic activity is constrained, stabilized, and made intelligible.

1.12 Novelty

This project advances a structural hypothesis about linguistic organization: that what is often treated metaphorically as a “linguistic space” can be specified, for analytic purposes, as a **multi-level generative geometry**—a constrained field of hosting positions and compositional interfaces that regulate how linguistic material becomes discourse-stable and combinatorially available. The empirical entry point is the **Reverse Entailment Diagnostic (RED)**, which operationalizes *nounal potential* by testing whether diverse linguistic material can be witnessed into discourse-addressable objecthood and then reverse-entails that capacity back to the base form. On the present proposal, the cumulative success and graded cost profile of RED motivates an architectural conclusion: the Universal Linguistic Domain is not invoked as an explanatory metaphor, but as the **hosting geometry required to make RED’s objectification facts systematic**.

Within that architecture, the Universal Linguistic Domain is modeled as an explicitly ordered set of levels: (i) a **Level-0 entry condition** (exeme insertion into an initially uncommitted hosting field), (ii) a **Level-1 hosting unit** (a term-level configuration with an internal Ordinate/Subordinate/Coordinate partition), (iii) a **Level-2 nucleation mechanism** in which coupling of term-level units yields a minimal predicate–argument skeleton (PAS) that supports the derivation of familiar syntactic objects (phrases, clauses, sentences, and certain compounds), (iv) a **Level-3 extension space** for higher-arity (four-term) supra-syntactical configurations, and (v) a **Level-4 discourse organization** in which chains of derived units are shaped by coherence, informational

management, and rhetorical constraints. The contribution is thus a cautious but testable claim: **RED provides a methodologically explicit probe of objectification**, and the proposed multi-level hosting geometry provides a compact way to model how that objectification is constrained and iterated from minimal hosting through syntax to text-level organization, without treating traditional category primitives as the primary explanatory units.

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Chapter 2 — Background and Existing Approaches

This chapter situates the Reverse Entailment Diagnostic (RED) within the existing landscape of theorizing about (i) grammatical categories, (ii) nominalization and “objectification” phenomena, and (iii) the syntax–semantics–discourse interfaces that make category-shifts both possible and constrained. Although RED is introduced as a bidirectional diagnostic for *nominal potential*—the capacity of linguistic material to behave as a syntactical noun entity in noun-slot positions —its significance becomes clearer when we see what earlier frameworks explain well, what they treat as peripheral, and where they typically stop short of an explicit, testable mechanism.

The key empirical pressure point is familiar: languages repeatedly allow actions, properties, relations, and even function-words to be stabilized as entity-like discourse objects (e.g., via genitives, determiner anchoring, nominalizers, clausal packaging, and quotation/mention). RED’s guiding idea is that these are not merely isolated “conversions,” but systematic witness environments that reveal an underlying, general capacity for

noun-slot hosting. What remains open—prior to the ULD formalization developed later—is how to state the mechanism of that capacity in a way that is (a) cross-linguistically portable, (b) cognitively plausible, and (c) computationally operational.

2.1 Historical overview of the problem

Classical and mid-20th-century generative approaches typically treat parts of speech as stable lexical categories, with nouns privileged as canonical arguments and verbs privileged as canonical predicates. The canonical statement of the nominalization problem in this lineage frames nominalization as a phenomenon to be derived from underlying structure and lexical properties (e.g., “Remarks on nominalization”; Chomsky 1970). In this view, category is not merely a convenient label; it is a grammatical primitive that constrains what can appear where.

Over time, however, two developments weakened the plausibility of a rigid “lexical essentialism.” First, morphosyntactic research accumulated a wide range of mixed-category and category-shifting constructions (conversion/zero-derivation, gerunds, nominalized clauses, predicative nominals, quotation/mention), pushing theorists toward finer distinctions between *roots*, *category heads*, and *constructional licensing*. Second, typological and cognitive work kept returning to the same descriptive generalization: speakers can reliably “make an entity” out of material that is not canonically nominal—often to support reference tracking, quantification, evaluation, or metadiscourse.

Contemporary frameworks respond to these pressures in different ways. Some preserve rigid category distinctions but multiply derivational devices; others weaken category primitives and relocate categorization to structures, features, or constructions; and still others treat “entityhood” as a discourse-semantic achievement rather than a lexical one. RED enters this history by making a methodological wager: instead of beginning with a category ontology and asking how a form changes category, begin with *witness constructions* that force entity-like behavior and ask what those witnesses entail backward about the base item’s noun-slot potential.

2.2 Major families of approaches

What follows groups existing work into several broad “families.” The goal is not to adjudicate every internal dispute, but to isolate the explanatory commitments that matter for RED and for the later ULD claim that nounal hosting reflects a universal slot-space.

2.2.1 Lexicalist and category-primitive approaches

A first family maintains that lexical items are typed for category (noun/verb/adjective, etc.), and that deviations are special operations: derivation, coercion, or metalinguistic mention. Within this perspective, the core question is often framed as: *what morphological or syntactic operation turns X into a noun?* The explanatory advantage is clarity: the grammar enforces strong constraints, and category-changing phenomena are localized to specific rules.

The cost is that many attested (and productive) patterns look less like isolated exceptions than like systematic exploitation of argument positions and discourse needs. When “noun-slot” environments are treated as merely a site for category conversion, the mechanism that makes noun-slot stabilization so generally available remains under-described.

2.2.2 Exoskeletal and root-based approaches

A second family weakens lexical category primitives by treating roots as category-neutral, with categorization emerging when a root combines with functional structure (e.g., Distributed Morphology). My RED book explicitly leverages this alignment: it notes that RED challenges rigid lexicalist category assumptions and is compatible with category-neutral roots and constructional licensing

This family explains, in a principled way, why the *same root* can surface as nominal or verbal, and why nominalizing morphology and syntactic packaging often behave as regular compositional devices rather than idiosyncratic “category flips.” But even here, a gap remains: “category neutrality of roots” does not yet specify *why* noun-slot stabilization is so broadly accessible across heterogeneous material (including items that are not easily treated as roots in the usual sense, such as function-words). RED takes the next step by treating witness success as the empirical basis for concluding *nounal reachability* rather than assuming categorical membership.

2.2.3 Construction-based and usage-based approaches

Construction Grammar and related usage-based frameworks relocate category behavior to the level of constructions: a form’s apparent category is what a construction makes it do (Goldberg 1995). My RED book explicitly uses this logic, noting that constructions like genitives and clausal subject frames can assign nominal roles regardless of the base’s prior category.

This family is particularly strong at describing how discourse tasks (reference, evaluation, topicalization, metacommentary) motivate objectification. Its recurring weakness, for present purposes, is methodological: if “the construction does it,” the theory still needs a disciplined test that distinguishes (i) robust objectification from (ii) marginal metalinguistic mention, stylistic play, or isolated coercions. RED is designed to be precisely such a discipline: a controlled set of witness environments with an explicit success criterion (entity-like behavior under noun-slot diagnostics).

2.2.4 Formal semantic approaches: type-shifting and nominalization

Formal semantics has long treated category shifts as type shifts, coercions, or shifts in semantic composition, often with special attention to event nominalizations, property nominalizations, and the semantics of genitives (e.g., Partee 1997; Dowty 1991). My RED book anchors its exposition in this literature by contrasting unidirectional entailment with a bidirectional diagnostic logic.

From RED’s viewpoint, the central contribution of this family is the insistence that “being noun-like” is not only about surface form; it is about entering a compositional and referential regime: quantification, reference, predication, possession, and discourse tracking. The limitation is that semantic type-shifts often get posited as *solutions* rather than *diagnostics*: they tell us how an interpretation can be built, but they do not by themselves give a testable procedure for when a base item’s noun-slot potential is empirically warranted.

2.2.5 Discourse and cognition: reference, addressability, and objecthood

A further family treats objectification as fundamentally tied to discourse management: establishing referents, maintaining accessibility, anchoring anaphora, structuring coherence (e.g., Heim 1982; Kamp & Reyle 1993; Ariel 1990; Gundel et al. 1993; Grosz et al. 1995). This family matters because it helps define what “noun-like entity” ultimately *means*: not “noun morphology,” but *addressable objecthood* in discourse.

The ULD materials I provided make this point explicit: witness success is evaluated by whether the output behaves as a discourse object, and languages differ mainly in which witness environments are economical, not in the success criterion itself. This perspective foreshadows the later claim that a universal noun slot is not merely syntactic but a structural condition on the space of discourse positions.

2.2.6 Contextuality and “quantum-inspired” perspectives

A recent family of proposals models contextual stabilization as selection among potential profiles under contextual “measurement” conditions. Independently of any commitment to the mathematics of quantum probability, the important shared intuition is that interpretive status is not fixed by bare form alone but is stabilized by structured contexts (frames, tasks, positions). This is relevant because it mirrors the empirical shape of the noun-slot puzzle: the same material can behave as an event, a property, or an entity depending on how it is hosted.

Where RED differs is that it does not merely restate contextuality; it insists on *witness design*: explicit environments that force stabilization and thus permit backward inference. In other words, RED turns “context matters” into “here are controlled contexts that entail back to nominal potential.”

2.3 Methods, data, and evidence types used so far

Across these families, four evidence types recur.

Introspective judgments and constructed examples. These are indispensable for isolating contrasts, but they are also vulnerable to (i) metalinguistic contamination, (ii) register effects, and (iii) unclear baselines for marginality. RED explicitly incorporates this by allowing graded interpretability (including marginal cases) but requiring that marginality be treated as a measurable cost rather than dismissed as noise.

Corpus evidence. Corpora reveal distributional regularities of packaging and noun-slot behavior (e.g., genitive environments, nominalizers, clausal subjects). My RED book notes corpus testing and evaluation against resources such as Universal Dependencies and large corpora. Corpus methods help distinguish productive constructions from one-off play, but they require annotation choices that can obscure the very category flexibility under investigation.

Psycholinguistic and neurolinguistic evidence. Modern neurocognitive models often distinguish stored lexical memory from combinatory unification/control mechanisms (e.g., Friederici 2011; Hagoort 2013). The RED book already positions the universal noun slot as cognitively grounded and cites neurological convergence as supporting evidence. This matters because noun-slot stabilization is plausibly a *process*, not merely a static representational fact.

Computational evaluation. NLP pipelines operationalize category assumptions through POS tagging, parsing, and semantic role labeling; they therefore provide a natural testbed for whether RED-like packaging detection improves downstream structure. My RED book explicitly frames parsing and SRL as target applications and reports pilot gains in subject detection and multilingual performance. Computational evaluation is uniquely valuable because it forces explicit criteria and failure modes.

2.4 Key debates, tensions, and unresolved questions

Several unresolved tensions motivate the need for a RED-based architecture that culminates in ULD.

Stability vs. flexibility. If categories are stable primitives, why does noun-slot stabilization seem so generally available? If everything is flexible, what prevents triviality?

Use vs. mention. Many theories acknowledge metalinguistic mention, but they rarely provide a principled way to separate “mere mention” from discourse-relevant objectification. RED’s reliance on witness environments and discourse-object behavior is intended to impose that discipline

Cross-linguistic comparability. Languages vary in their packaging devices (genitives, nominalizers, determiners, clause structure). The challenge is to define a universal test whose success criterion is stable while the witness inventory is language-specific. The ULD materials state this explicitly as a design principle.

Gradience and cost. Acceptability and productivity are often graded. A binary category ontology is poorly matched to this. RED’s later development treats reachability as metric (cost) rather than categorical membership

Mechanism vs. label. “Universal noun slot” can be used as a descriptive label, but the field needs a mechanism: a structured account of what it means for something to be “hosted” as an entity.

2.5 Where RED fits, and why it points beyond itself toward ULD

RED is continuous with earlier work in at least three ways: it agrees that constructions and functional structure matter (CG/DM), it agrees that noun-like behavior is ultimately about reference and discourse objecthood (discourse semantics), and it agrees that contextual conditions select interpretations (contextuality accounts). My RED book explicitly positions RED against rigid lexicalist frameworks while aligning it with more dynamic models

RED is discontinuous in two decisive respects.

First, it is not merely a theory of how nominalizations are formed; it is a *diagnostic methodology* that uses controlled derivational and positional witnesses to support backward inference to nominal potential. In the book's terms: RED does not ask "what category is X?"; it asks whether X can be *hosted* in a nounal position and what packaging it requires to be interpretable.

Second, and more importantly for the architecture of this monograph, sustained success of RED-witnesses forces an inference about structure. If objectification is available across the full range of targetable material, then there must be an intrinsic space of admissible hosting positions in which such stabilization can occur. The ULD materials state this as an "architectural inference": systematic witness success motivates identifying a universal domain of linguistic activity with a universal objecthood domain (ULD $\equiv$ UNS).

This is the conceptual transition from RED to ULD:

- **RED** supplies empirically controllable *witness environments* and a success criterion (discourse-object behavior under noun-slot diagnostics).
- Repeated witness success across categories and languages motivates the claim that "noun-slot" is not an accidental corner of grammar, but a **structural region of a universal hosting space**.
- **ULD** is introduced as that hosting space: a constrained geometry of possible discourse positions within which exeme-level material can be stabilized, combined, and chained into higher units.

In short, RED is the empirical entry point; ULD is the explanatory space that makes RED's generality intelligible as a consequence of domain structure rather than as a collection of exceptions.

This bridge also clarifies why cognitive science and computational applications are not "addons." If RED is about controlled stabilization into objecthood, then (i) neurocognitive models of unification/control predict processing differences for higher-cost objectifications, and (ii) computational pipelines should benefit when witness environments are used to supervise promotion of event/property material into argumental nodes for parsing and SRL.

2.5.1 Positioning: Where RED–ULD Fits in the Landscape

The framework developed in this book is best located at the intersection of two established "neighborhoods" in contemporary theory: **exoskeletal category formation** and **construction-based licensing**. The closest intellectual kin are **Distributed Morphology** and related root-based approaches (Halle & Marantz, 1993), together with **Construction Grammar** and broader constructional/usage-based accounts (Goldberg, 1995). These traditions already share two commitments that RED makes explicit and operational: first, that category behavior is not exhausted by lexical labels; and second, that structural environments can force stable interpretive outcomes.

However, the present book is not simply an application of DM or Construction Grammar to nominalization. It advances a stronger architectural claim. It proposes that the recurrent success of nominal stabilization across categories points to a **Universal Linguistic Domain (ULD)**: a constrained space of admissible discourse-positions in which linguistic material can be stabilized as entity-like, predicate-like, or otherwise. On this view, noun-slot stabilization is not merely a family of constructions, nor merely a feature-checking option; it is a privileged region of possible discourse—one that grammars and minds reliably exploit for reference, possession, quantification, and the treatment of actions or properties as addressable "things."

Within that architecture, **RED** is not presented as a collection of clever examples. It is a disciplined evidential method: a way of determining when a form’s behavior warrants the inference that it can be hosted as an entity in noun-slot environments, and of mapping the conditions under which that hosting is easy, costly, marginal, or constructionally restricted.

2.6 Critical comparison: What the closest theories get right—and where ULD departs

Distributed Morphology (DM) gets the direction exactly right: category is treated as a **structural outcome** rather than as an inherent lexical essence (Halle & Marantz, 1993). This aligns directly with RED’s foundational stance: what matters is not what a unit “is” in isolation, but what it can be licensed to do under specific hosting conditions. Where ULD departs is the level of explanation. DM supplies a powerful internal mechanism for categorization via functional structure, but it does not, by itself, articulate a **single domain geometry** in which noun-slot stabilization is a privileged *region of possible discourse* rather than one more configuration among many. ULD makes that domain explicit and treats “noun-slot availability” as a structural property of the discourse-space itself.

Construction Grammar (CxG) also aligns closely with RED’s empirical posture: constructions can force entity-like behavior by placing material into frames that require reference and argumenthood (Goldberg, 1995). CxG is especially strong at describing how discourse tasks motivate objectification and packaging. Where ULD departs is methodological and architectural. Construction-based explanations can risk becoming descriptive catalogues unless they are paired with a disciplined test that separates robust objectification from mere rhetorical play, marginal mention, or idiosyncratic coercion. RED supplies that discipline: it treats nominal environments as controlled witnesses and makes backward inference part of the method. ULD then explains why those witnesses are systematic: they succeed because the domain contains a stable noun-slot region that constructions can reliably target.

Formal semantics of argument structure and nominal regimes provides indispensable tools for stating what “noun-like behavior” amounts to in compositional terms—especially in the organization of predicate–argument relations and the semantics of genitives and nominalizations (Dowty, 1991; Partee, 1997). Where RED–ULD departs is the direction of reasoning. Much formal work explains how nominal readings are built (type shifts, coercion operators, composition rules), but RED treats nominal outputs as **evidence**: if the grammar repeatedly licenses nominal packaging, then we can infer that noun-slot hosting is a general capacity rather than a narrow constructional exception. ULD generalizes this further: it treats such licensing as a window into the structure of the domain of discourse positions itself.

Cognitive and discourse-oriented approaches correctly locate objectification in the management of reference, addressability, and discourse coherence—why we turn actions and properties into “things” that can be tracked, evaluated, and related to other things (Lakoff & Johnson, 1980). ULD is continuous with this perspective, but it refuses to leave “reification” as a metaphor. It claims that reification corresponds to a structured stabilization regime: a principled kind of hosting in the universal discourse-space, with predictable costs, constraints, and cross-linguistic surface strategies.

Finally, the computational perspective clarifies what is at stake. Many NLP pipelines treat category labels as stable upstream facts, and their failure modes often concentrate exactly where RED predicts: clausal subjects, nominalized events, complex nominals, and category-shifting constructions. A RED-guided pipeline reframes these not as “noise,” but as diagnostically privileged sites for improving parsing and semantic role labeling—especially where argument structure must be recovered under heavy packaging and cross-linguistic variation (Dowty, 1991; Palmer et al., 2005).

In short, DM supplies category-by-structure, CxG supplies constructional licensing, formal semantics supplies compositional constraints, and cognitive/discourse approaches supply the motivations for objecthood. What none of them states explicitly, in the form required by this book, is the unifying claim: a **Universal Linguistic Domain as the constrained space of possible discourse-positions**, with a **universal noun-slot region** as a

privileged stabilization site, and **RED** as the diagnostic instrument that gives empirical access to that geometry.

2.7 Chapter summary and bridge to Chapter 3

This chapter has mapped the main ways existing work approaches category behavior and nominal stabilization: lexicalist and category-primitive traditions, exoskeletal and root-based models, constructional and usage-based accounts, formal semantic treatments of argument structure and nominal regimes, and cognitive/discourse perspectives on objectification. Each explains part of the phenomenon, but each tends to leave at least one of the following underspecified: a cross-linguistically portable diagnostic logic, a principled separation between robust noun-slot stabilization and mere mention, and a structural account of why noun-slot hosting is so general without collapsing categories into triviality.

These gaps motivate the next step. Chapter 3 introduces the Universal Linguistic Domain explicitly as a constrained geometry of discourse-positions and shows how RED functions within that geometry: not as a floating heuristic, but as an evidential probe that reveals when and how linguistic material enters the noun-slot region of the domain.

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Chapter 3 — Core Theory: The Universal Linguistic Domain

Chapter 2 ended by motivating a shift in explanatory level: RED cannot remain “just a diagnostic,” because its systematic success across witnesses and languages forces an architectural question—*what kind of grammatical space makes that success possible?*

In the Chapter-2 bridge, this is stated explicitly as the transition from RED to ULD: repeated witness success motivates treating “noun-slot” not as an accidental corner of grammar, but as a stable region inside a universal hosting space.

This chapter introduces that hosting space. The claim is not merely that speakers *can* nominalize many things, or that constructions *sometimes* override category expectations (as in Construction Grammar), or that category can be assigned by syntactic structure (as in Distributed Morphology). Those are close intellectual neighbors.

The stronger claim is that linguistic activity itself unfolds within a constrained “geometry” of admissible discourse-positions—positions that are cognitively targetable, grammatically licensable, and structurally repeatable across languages. Chapter 2 already positioned this as the point where ULD departs from DM/CxG: it makes the *domain* explicit and treats noun-slot availability as a property of the discourse-space rather than as one more constructional option among many.

3.1 The transition from RED to ULD

RED begins with a methodological wager: instead of starting from a fixed category ontology (“verbs are predicates; nouns are arguments”) and then inventing special operations for category-shift, we begin with controlled *witness environments* that force entity-like behavior, and we infer backward from success to *nominal potential*.

But once you take that inferential direction seriously, you immediately face an explanatory constraint. If objectification is reliably available—if actions, properties, relations, and even function-words can be made into discourse objects under systematic conditions—then “nounhood” is not behaving like a narrow lexical class. It is behaving like a *reachable hosting region* that the grammar can repeatedly target. Chapter 2 formulates the resulting move as an architectural inference: systematic witness success motivates identifying a universal domain of linguistic activity whose internal structure includes an objecthood region (ULD $\equiv$ UNS).

So the RED→ULD transition is not a change of topic. It is a tightening of explanatory responsibility:

- RED supplies the test: witness environments + noun-slot behavior + reverse entailment.
- ULD supplies the reason the test can generalize: noun-slot is a structural region of a universal hosting space.
- Cognitive science and computation are therefore not “add-ons,” because if the phenomenon is a *general stabilization interface*, it should show processing signatures and it should improve formal pipelines that currently fail exactly where category is not “as expected.”

3.2 The ULD axiom: ULD = UNS

The central axiom is stated in the ULD materials as follows: “Axiom (ULD = UNS). The Universal Linguistic Domain is identical to the Universal Nounal Slot.”

This is the strongest form of the universal noun-slot thesis. It says: the deepest invariant is not that nouns exist everywhere, but that *everything grammar can target is, in principle, targetable as a noun-like object*—a “thing” for the purposes of reference, re-entry, packaging, and discourse manipulation (even if sometimes only in high-cost, metalinguistic regimes).

Two immediate consequences follow.

First, the ULD is not a list of word classes or a bundle of separate modules. It is “one universal nounal slot space with internal positional geometry,” organized by levels of hosting and combination.

Second, once ULD = UNS, the empirical work is no longer carried by a binary question (“is X a noun?”). It is carried by *reachability and cost*: which stabilizations are low-cost and productive, which are medium-cost and construction-restricted, and which are high-cost and largely metalinguistic.

3.3 Levels of organization: from exeme entry to discourse space

The ULD’s “geometry” is easiest to grasp as a layered organization of hosting and composition.

Level-0: entry (exeme).

At the entry point, the basic unit is an *exeme*: a modality-neutral expression unit—a form-pattern considered as expression, independent of grammatical category, lexical status, or meaning.

This is the minimal “something” that can enter the hosting field.

Level-1: the one-term structure (the term).

The first linguistic organization is the *term* (t): a three-region hosting tape—Ordinate / Subordinate / Coordinate—providing the smallest internally articulated stabilizing frame for an item as a discourse-target.

At this level, the theory’s commitment is: there is no “free-floating word meaning” outside hosting; even the simplest usable item appears as a hosted term.

Level-2: the two-term structure (predication) and domain organization.

When two Level-1 terms are combined into a predication relation ($t_1 R t_2$), we obtain the earliest generative skeleton: a predicate–argument frame, a minimal combinatorial architecture that organizes roles and slotting prior to fully elaborated sentence forms (Dowty 1991; Halle & Marantz 1993). This is the “nucleating layer” in which predicate–argument structure becomes available as a structured object rather than as a descriptive afterthought. Chapter 2 explicitly describes this as the level at which ULD is treated as a generative geometry linking predicate–argument structure to sentence formation and discourse chaining.

At this point we can state the book’s crucial reclassification: what traditional grammar treats as primitive—sentences, clauses, many phrase types, even certain compounds—are not primitive here. They are *derived* elaborations of Level-2 skeletons: embeddings, extensions, and reconfigurations of the predicate–argument frame into increasingly articulated units (cf. Goldberg 1995 on argument-structure constructions; and classic work on nominalization and functional structure).

Level-3: supra-syntactical combination (four-term structures).

Two Level-2 units can be combined into higher-complexity configurations ($2 \times (t_1 R t_2)$), yielding supra-syntactical structures: coordination complexes, operator-scopal architectures, extended modification chains, and other “text-functional” units that begin to behave like building blocks of discourse organization.

Level-4: text (domain of discourse forces).

Finally, chains (or circuits) of derived units form *text*: a higher-order domain in which topic management,

rhetorical contrast, evaluation, genre constraints, and coherence pressures operate over and through syntactic structure without reducing to it. This is still “inside” ULD: the claim is that discourse forces bias which slots are foregrounded, which entity-profiles are stabilized, and how predicate–argument skeletons are chained into extended interpretation.

These levels are the ULD’s dimensions: together they constitute the “area of possible linguistic discourse”—the structured space within which linguistic activity becomes possible at all.

3.4 What, exactly, is being diagnosed: exeme, lexeme, lemma

RED cannot be clean unless we separate (i) expression form, (ii) lexical organization, and (iii) indexing conventions—because debates about “whether a verb can be a noun” often slide between these layers without noticing.

- **Exeme:** the pre-categorical form-pattern (modality-neutral expression unit).
- **Lexeme:** the language-specific lexical unit organized over one or more exeme realizations, carrying structured semantic and distributional potential.
- **Lemma:** the conventional citation form used to index a lexeme in dictionaries, corpora, and annotation systems; an external mapping key, not the lexical object itself.

Once this is in place, the RED→ULD connection becomes precise: RED is not “turning verbs into nouns.” It is testing whether a given exeme/lexeme, under a controlled witness strategy, can be stabilized in noun-slot regimes—i.e., can behave as a discourse object under the diagnostics that define noun-slot behavior (reference, possession/genitive compatibility, anaphoric re-entry, etc.).

3.5 Exeme, Persistence, and the Universal Domain

3.5.1 Exeme: Entry Unit

A central design choice of this book is to separate **what enters the system** from **what the system later stabilizes as a lexical object**. The entry unit is what I call an **exeme**: a modality-neutral expression pattern treated, at the moment of entry, purely as **form**.

Definition (Exeme). An **exeme** is a modality-neutral expression unit: an abstract **pattern of form** (e.g., orthographic *r-u-n*, phonological *[ran]*, or any conventionalized encoding) considered independently of grammatical category, lexical status, or meaning. It is the **lowest formal layer**—what enters the hosting field at **Level-0**.

This definition is intentionally strict. In ULD/UNS, an exeme does not *arrive* as a noun, verb, adjective, or function word; it arrives as a candidate for hosting. Category effects arise only after placement. In other words, the model treats **category as a hosting outcome**, not as an intrinsic property of what is inserted.

This is also why Level-0 matters architecturally: the ULD hypothesis claims that grammatical work is done by the geometry of available positions. If so, then the theory needs a minimal unit that can be placed **without already carrying** the very categories the geometry is supposed to explain. Level-0 provides that unit.

What an exeme is not

An exeme is **not** a token in a specific sentence or recording; it is not “this particular written occurrence” or “this particular acoustic event.” It is also **not** yet a lexeme (a language-internal lexical object) and **not** a lemma (a citation label used by dictionaries and corpora). Those are later stabilizations and conventions built over the entry layer.

Because Level-0 is only form, we need a principled way to state how form becomes “a word” in a language, and how “a word” becomes indexable in resources.

- **Exeme** = form-pattern (pre-categorical input layer).
- **Lexeme** = a language-internal lexical object: exeme(s) organized into structured meaning plus a grammar/hosting signature (what profiles it can take, and where).
- **Lemma** = an indexing convention: a canonical citation key used by lexicography/corpora to refer to a lexeme (often many-to-many in practice).

This triad is not bookkeeping; it is what lets the book keep three issues separate that are often conflated: **form identity**, **lexical organization**, and **reference/annotation practice**.

Why exemes connect RED to ULD (the inferential direction)

RED is introduced as an empirical probe: it tests whether diverse material can be stabilized into noun-slot behavior under nominal environments, and then reverse-entails that capacity back to the base. The move from RED to ULD becomes conceptually forced once we recognize that RED’s successes presuppose a background hosting space in which “becoming an entity” is a *positionally licensed event*.

Exemes are the Level-0 points where this becomes explicit. We often encounter language through **lemma- and lexeme-facing evidence** (dictionary headwords, corpus annotations, paradigms, derivational families). But if a lemma is well-defined as an index, that already supports a backward inference that there exists a determinate **exeme-level form pattern** being indexed.

Likewise, when derivations and distributions are licensed only under particular category environments, their acceptability supports a reverse inference that an underlying exeme exists and can be hosted into the relevant profile under the relevant construction.

So, in the book’s logic:

1. **ULD** supplies the geometry of admissible hosting positions.
2. **Exemes** supply the minimal insertable material that can be hosted *without presupposing category*.
3. **RED** supplies the observable tests that reveal when hosting has stabilized an item into noun-slot behavior, and thereby lets us infer constraints on that geometry.

A note on cognitive and computational alignment

Because exemes are defined at the form layer, they provide a clean interface to (i) perceptual encodings across modalities (orthographic, phonological, etc.) and (ii) NLP preprocessing decisions (tokenization, surface normalization), while keeping lexical and inferential commitments (lexeme structure; lemma indexing; RED inferences) analytically distinct. This separation is what makes later chapters’ cognitive and computational claims stateable without collapsing “what the system hosts” into “how a corpus labels it.”

3.5.2 Persistence, and the Universal Domain

A diagnostic like RED only becomes explanatory once we are clear about *what kind of object* it is diagnosing. The book therefore distinguishes three layers that are often collapsed in ordinary linguistic practice and in computational preprocessing: **exeme**, **lexeme**, and **lemma**. A **lemma** is an indexing handle (a citation form or database key); a **lexeme** is a language-specific lexical unit (an organized cluster of distributional/semantic potential); and an **exeme** is the lowest-level formal unit: a modality-neutral *expression pattern* that can be realized in writing, speech, signing, or other media.

On the present view, an **exeme** is not “a noun” or “a verb.” It is simply an entry unit that can be hosted in different positions and thereby acquire different profiles.

In the earlier formulation, an exeme is defined as a “specific sequence of expression considered purely as a pattern of form,” independent of grammatical category—e.g., the same underlying pattern may be instantiated as written **r-u-n**, spoken [rʌn], and so on.

This matters because RED’s evidence is typically gathered from *surface* objects (tokens, orthographic strings, conventional lemmas), but the theory’s claims are about *stability across contexts, media, and constructions*. Without an exeme layer, we cannot even state what stays constant when we say “the same material” shifts between predicate-like and entity-like behavior.

This is where the **lexical ontology persistence principle** enters. The persistence claim is not that meaning is context-free; rather, it is that a lexical object has an internal architecture that remains identifiable across contextual deployments, while context operates as a *profiling* (or selection) function over that architecture. In the book’s earlier statement: the identity of a lexical object is the persistence of its architecture across uses, and “context becomes a profiling operation.”

Conceptually, the exeme provides the *formal anchor* for tracking “the same thing” through variation of medium and token noise, while the lexeme provides the *structured lexical identity* whose architecture is being profiled rather than recreated.

RED then supplies the bridge from these ontological commitments to empirical work. RED begins with what I called the “road signals”: derived nominalizations, possessives, clausal packagers, and other witness environments that force an item to behave as an addressable discourse object. From a successful witness (an output that occupies noun-slot positions and supports reference maintenance), RED reverse-entails back to the base and infers **nominal potential**—not as a stipulation, but as a property revealed under controlled hosting conditions.

Crucially, that inference can be stated as a ladder: from **noun-slot witness** → to **lemma-level** behavior → to **lexeme-level** capacity → to **exeme-level** availability under hosting.

But once RED is taken seriously as *cross-category* and *cross-linguistic* (and once witness success is systematic rather than anecdotal), the diagnostic forces a further theoretical step: there must exist a background **space of admissible hosting positions** in which objectification is possible at all. This is the transition from **RED to ULD**. In my own formulation, ULD is “one universal nounal slot space with internal positional geometry,” where **Level-0** is exeme entry into an empty hosting field, **Level-1** is the one-term “tape” (Ordinate/Subordinate/Coordinate), and **Level-2** is the two-term / three-domain organization (S–P–O) with the special predicate connector and micro-slots; RED is the empirical method that demonstrates universal nounal hosting, while ULD (= UNS) is the theoretical structure that explains how that hosting is organized and constrained.

In other words: **RED discovers the universality of noun-slot access; ULD formalizes the geometry that makes such access a coherent architectural property rather than a grab-bag of constructional tricks.**

This RED→ULD move also clarifies why the exeme is the correct entry point for Chapter 3’s core theory. If ULD is the “field of possible discourse,” then the most fundamental unit that *enters* that field must be defined below lexical category labels; it must be something that can be hosted *as* noun-like, verb-like, etc. That is exactly what the exeme is: a category-neutral formal pattern whose profiles are stabilized by where and how it is hosted.

Finally, this architecture is not merely philosophical—it yields interface predictions and applications. Neurocognitively, the universal noun-slot story predicts convergences when event/property material is “objectified,” consistent with findings that nominalized forms recruit both object-related and action-related circuitry and that noun phrases recruit unification resources (e.g., IFG) under combinatorial demands.

Computationally, witness environments are detectable corpus patterns that can supervise “promotion” of event/property material into argumental nodes, improving parsing and role labeling when packaging is present. In my own pilot formulation, a RED-enhanced pipeline is reported to improve subject detection over a baseline parser and to yield measurable gains for SRL in nominalized configurations.

Put compactly: **the exeme is the entry-unit; persistence is the claim about what remains stable across profiling; RED is the evidential lever that reveals noun-slot availability; and ULD is the geometric theory of the hosting space in which that availability becomes structurally intelligible.**

3.6 RED restated inside ULD = UNS: from “nounhood” to reachability and cost

In a noun-centered ULD, RED must be stated carefully. If ULD = UNS, then RED is no longer “a path into nounhood,” because the domain itself is nounal. Instead, RED becomes an *edge-probe*: it detects family membership and low/medium/high-cost reachability inside UNS.

This is why the ULD notes insist: “Cost, not membership, carries the empirical load.”

The diagnostic logic stays the same (witness → noun-slot behavior → reverse entailment), but the interpretation changes: what varies is not whether an item “is in the domain,” but how economically it can be stabilized as an object under available witnessing strategies.

The worked derivations make the point concretely:

- **Low cost:** *run* → *a run* (productive determiner anchoring; easy anaphora; ordinary possessives).
- **Medium cost:** *red* → *the red* (restricted reading; construction-dependent).
- **High cost:** *and* → *the “and”* (quotation/mention regime; typically metalinguistic).

These cases show what ULD is buying: a single domain with internal geometry, where differences in “category behavior” are modeled as differences in transition constraints and graded costs—not as exceptions or mere rhetorical flourishes.

3.7 The geometry is constraint, not metaphor

Calling ULD a “geometry” is not decorative. The theory requires constraints that define *possible paths*, *barriers*, and *graded transitions*, rather than merely listing permissible word classes. The ULD notes state these constraints explicitly—type compatibility, recoverability/distinctness, scope well-formedness, discourse accessibility, and economy/minimal packaging.

This is also where the mechanism behind “context” becomes tractable. “Context” is not a mystical force; it is a biasing field that pushes hosting choices toward certain regions of the domain (e.g., toward objecthood positions when reference tracking, evaluation, quotation, or possession becomes discourse-relevant). The ULD is the structure that makes those biases intelligible and testable.

3.8 Cognitive and computational consequences are built in, not appended

Chapter 2 already framed this point bluntly: if RED is about controlled stabilization into objecthood, then cognitive science and NLP are not optional decorations—they are natural testbeds.

Cognitive / neurolinguistic interface.

If stabilization into noun-slot regimes is a general interface operation, then higher-cost objectifications should recruit more effortful integration/unification control, and low-cost objectifications should pattern more like ordinary referential processing. Chapter 2 aligns this with neurocognitive models that distinguish lexical memory from combinatory unification/control mechanisms (e.g., Friederici 2011; Hagoort 2013).

The prediction is not “a noun module,” but systematic processing signatures for the *operation* of packaging predicate-like material into objecthood configurations.

Computational interface (parsing and SRL).

Dependency parsing and semantic role labeling often fail where arguments are realized through nominalized structures, clausal subjects, complex nominals, and other category-shifting regimes. Chapter 2 states this as a direct motivation for computational evaluation: NLP pipelines operationalize category assumptions, so they provide an unusually strict test of whether RED-style packaging detection improves downstream structure.

On the ULD view, a RED-supervised component is not “extra linguistics”: it is an explicit procedure for promoting event/property material into argumental nodes when witness environments license such promotion, thereby improving PAS recovery and role assignment.

(Chapters 8–10 will return to these consequences in depth; the present chapter’s point is conceptual: ULD makes these consequences *predictable* because stabilization is treated as a domain-level operation with measurable costs, not as an idiosyncratic trick.)

3.9 Where we are, and what comes next

We can now state the book’s core theoretical package in one line.

RED is the empirical entry point: a disciplined method for testing when linguistic material behaves as a discourse object under controlled noun-slot witnesses.

ULD is the explanatory space: the constrained geometry of possible discourse-positions within which such stabilization is possible and systematic, with noun-slot availability treated as a privileged region of that space rather than as a scattered family of exceptions.

The next chapters will tighten the formal side (making the level-structure and slot topology fully explicit), then expand the empirical side (cross-linguistic witnessing and boundary cases), and finally return to the cognitive and computational consequences with implementable criteria.

Chapter 3 References (short list; consistent with Chapter 2/Book bibliography)

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Chapter 4 — Detailed Analysis: Data and Case Studies

4.1 Aim of this chapter

Chapter 3 proposed a simple but demanding claim: *entity-stabilization is not an accidental trick of a few constructions, but a systematic capacity of grammar and discourse*. The purpose of the present chapter is to *make that claim earn its keep* by slowing down and working through concrete data in a way that shows (i) what counts as evidence, (ii) how the Reverse Entailment Diagnostic (RED) is applied step by step, and (iii) why the recurring success of these tests is best explained by a domain-level architecture—ULD as a constrained space of admissible discourse positions—rather than by a list of isolated exceptions. This chapter therefore follows the case-study logic recommended by the book’s own data-chapter template: each case study states a question, presents a small dataset, runs the analysis explicitly, and closes with interim conclusions that feed the cross-case synthesis.

A second goal is to keep the empirical story aligned with the corrected theoretical picture developed in the ULD materials: if ULD is identified with a universal objecthood space, then RED is best treated not as “a path into nounhood,” but as an **edge-probe** that tests *low-/moderate-cost reachability* into the noun-slot region under controlled witness environments.

Put plainly: the data here are not just “examples of nominalization.” They are *witnesses* for how a linguistic system stabilizes material as a discourse object, and for how expensive (or cheap) that stabilization is across languages and constructions.

4.2 Data sources and evidence types

The empirical backbone of the chapter is deliberately mixed, because the phenomenon itself is mixed: some objectifications are ordinary and productive; others are marginal and strongly metalinguistic. Accordingly, I draw on three evidence types.

(i) Controlled constructed minimal pairs. These are the primary didactic engine of the chapter: they are designed to isolate the relevant diagnostic environments (possessives/genitives, subject/object placement, predicate nominal positions) while holding other factors constant.

(ii) Cross-linguistic marker contrasts. The same witness logic is run across English, Mandarin, Tongan, Spanish, and Arabic, with particular attention to how each language packages “entity-like” behavior (e.g., genitive/possessive strategies, nominalizers, clause-to-NP options).

(iii) Corpus checks and judgment scaffolding. Where available, corpora are used to verify that a pattern is attested and not merely imaginable; where corpora are thin (e.g., low-resource cases), native-speaker judgments play a larger role. The originating RED manuscript explicitly reports using COCA, CCL, and Universal Dependencies alongside native judgments for Tongan and Arabic, and it notes that marginal forms were evaluated in metalinguistic contexts.

Because the point of this chapter is *diagnostic clarity* rather than maximal typological coverage, the datasets are intentionally small but explanatory: each case is chosen to expose a distinct “slice of reality” (ordinary predication, event-to-entity packaging, metalinguistic objectification), consistent with the book’s own design principle for Chapter 4.

Notation note (lightweight glossing). Interlinear glosses are schematic and focus only on the diagnostic morpheme(s) that realize packaging into an entity profile (e.g., GEN/POSS markers such as Mandarin *de*, Tongan *’a*, Spanish *de*, Arabic genitive morphology). Full morphosyntactic segmentation is not attempted here; the aim is to keep the RED witness environment visible.

4.3 Method: RED as witness logic under a domain hypothesis

RED, in its operational form, proceeds as follows: apply a nominal packaging operation (overt or covert), test interpretability/grammaticality, and then reverse-entail back to the base to confirm noun-slot potential by actual placement in subject/object environments.

What matters for the present chapter is how this diagnostic is *reinterpreted* once we accept the architectural transition from RED to ULD:

Repeated witness success across categories and languages motivates treating noun-slot stabilization not as a corner case, but as evidence for a universal hosting space—a constrained geometry of discourse positions within which even exeme-level material can be stabilized and chained into higher units.

This also motivates a crucial refinement: **strength is not binary; it is metric**. In the noun-centered ULD framing, the discriminating variable is not “nounhood vs. non-nounhood,” but **cost**—how cheaply and robustly an item traverses from its active functional family into an entity-hosting configuration. Low cost corresponds to strong, productive nominal potential; high cost corresponds to marked, metalinguistic objectification.

In the analyses that follow, I therefore use a three-way descriptive scale:

- **RED-strong:** productive, ordinary discourse objectification (low cost).
 - **RED-mid:** grammatically available but constructionally constrained or register-biased (moderate cost).
 - **RED-weak:** primarily metalinguistic or borderline; interpretability depends on explicit mention contexts (high cost).
-

4.4 Case Study 1 — Property/kind material stabilized as an entity (“square”)

4.4.1 Context and question

Many theories readily grant that *nouns* occupy noun slots. The first stress test is whether **property/kind material**—often treated as adjectival or predication—can be stabilized as an entity in a way that is systematic and cross-linguistic. The guiding question is:

When a form participates in property-like predication, what evidence shows that it nonetheless has a low-cost path into noun-slot behavior?

4.4.2 Dataset (five languages; genitive witness)

(4.1) English

- a. *Square is a quadrilateral.*
- b. *Square’s sides are equal.*

(4.2) Mandarin

- a. 正方形 是 四边形。
zhèngfāngxíng shì sìbiānxíng
‘A square is a quadrilateral.’
- b. 正方形的 边 ...
zhèngfāngxíng de biān ...
‘square GEN side(s) ...’

(4.3) Tongan

- a. *Fokika ...* ‘(A) square ...’ (predicational frame)
- b. *fokika ’a ...* ‘square POSS ...’

(4.4) Spanish

- a. *Un cuadrado es un cuadrilátero.*
- b. *Los lados del cuadrado ...* ('the sides of-the square ...')

(4.5) Arabic (schematic)

- a. 'square = quadrilateral' predicational frame
- b. 'sides of the square' with genitive marking on the dependent noun phrase

These are the same core contrasts used in the RED article's cross-linguistic evidence set: base predication plus a genitive/possessive environment that forces entity-packaging.

4.4.3 Analysis using the framework

The decisive witness here is not the mere existence of a possessive construction (all languages have *some* possessive strategy), but the *reverse-entailment inference* it licenses:

1. **Packaging:** the genitive/possessive configuration requires an entity-like host (something that can bear possession/relational anchoring; cf. Partee's treatment of genitives as diagnostics of nominal behavior).
2. **Witness success:** the derived form is interpretable and grammatical in ordinary discourse (*square's sides*, *zhèngfāngxíng de biān*, *del cuadrado*).
3. **Reverse entailment:** from the success of the derived configuration, we infer that the base item is licensed for noun-slot behavior (subject/object), even when it also participates in predicational frames.

In ULD terms, Case Study 1 illustrates a **low-cost traversal into the noun-slot region**: the language provides an ordinary packaging operation (genitive/possessive) that stabilizes the item as a discourse object. On the ULD=UNS axiom, this is expected: the domain is an objecthood space, and what varies is not membership but cost and licensing.

4.4.4 Interim conclusion

This is a RED-strong pattern: property/kind material can be stabilized as an entity under ordinary genitive witness environments across typologically diverse languages. The cost is low because the relevant constructions are productive, not exotic, and do not require explicit metalinguistic mention.

4.5 Case Study 2 — Event/property material stabilized as an entity ("to book carefully")

4.5.1 Context and question

The second slice of reality is more demanding: **eventive material** is often treated as "verbal," yet discourse routinely treats events as objects (advantages, reasons, priorities). The question is:

Can a clearly predicational base (an event/property structure) be forced into noun-slot behavior in a way that RED can certify—and can we compare the cost of that certification across languages?

4.5.2 Dataset (nominalization + genitive witness)

(4.6) English

- a. *To book carefully is my priority.*
- b. *To book carefully's advantage is clear.* (marginal for some speakers; improved under explicit mention contexts)
- c. *Booking carefully saves time.*

(4.7) Mandarin

a. 小心预订是优先事项。

xiǎoxīn yùdìng shì yōuxiān shìxiàng

‘Booking carefully is a priority.’

b. 小心预订的优点 ...

xiǎoxīn yùdìng de yōudiǎn ...

‘the advantage(s) of booking carefully ...’

(4.8) Spanish

a. *Reservar cuidadosamente es mi prioridad.*

b. *La ventaja de reservar cuidadosamente ...*

(4.9) Tongan / Arabic (schematic)

Parallel contrasts with possessive/genitive packaging and clausal-subject style noun-slot occupancy.

Again, these are drawn directly from the RED article’s multi-language “to book carefully” set, including the observation that some English genitive-over-clause cases are marginal but interpretable, especially metalinguistically.

4.5.3 Analysis using the framework

The key point of Case Study 2 is that **noun-slot occupancy and possessive packaging can dissociate**. Many speakers accept clausal subjects (*To book carefully is my priority*) more readily than genitive clausal hosts (*to book carefully’s advantage*). That asymmetry is not a problem for RED; it is a signal about **cost and licensing**:

- The clausal-subject configuration already forces an entity-profile (“something that can be prioritized, evaluated, compared”).
- The genitive configuration demands an additional kind of packaging—treating the entire eventive unit as a possessable anchor—which increases cost in some grammars.

ULD makes this pattern legible: the traversal into noun-slot stabilization is possible, but edges differ in weight. The chapter-level claim is therefore not “everything is equally nominal,” but rather: **entity-stabilization is universally available, while the costs of particular packaging routes are language- and construction-sensitive**.

4.5.4 Interim conclusion

Case Study 2 is typically RED-strong for clausal-subject objectification and RED-mid for genitive-over-event objectification (English especially). The important result is the *structured asymmetry*: the system does not collapse into triviality; it differentiates routes by cost.

4.6 Case Study 3 — Hard data: function words and explicit mention (“and”, “after”, “the”)

4.6.1 Context and question

The third slice of reality targets the standard objection: “My examples work only where there is rich lexical meaning. Surely function words are different.” The question is:

Can even grammatical machinery be stabilized as a discourse object—and if so, is that stabilization systematically higher-cost and more metalinguistic?

4.6.2 Dataset (edge cases; metalinguistic scaffold)

(4.10) Conjunction

a. *And* is a connector.

b. *And's* role in this sentence is contrastive. (marked; improves in pedagogical/metalinguistic registers)

(4.11) Preposition

a. *After* is uncertain.

b. *After's* consequences in the narrative are delayed. (marked; discourse-analytic register)

(4.12) Article

a. *The* is a determiner.

b. *The's* use here signals familiarity. (highly marked; mostly grammatical description)

These are the same “other parts of speech” demonstrations used in the RED article to show universality, with the explicit acknowledgment that article-based cases are marginal and depend heavily on mention contexts.

4.6.3 Analysis using the framework

This is precisely where the ULD restatement does its most useful work. Under ULD=UNS, the question is not “can *the* become a noun?” (a framing that invites ridicule), but:

What is the cost of objectifying grammatical machinery as a discourse target, and what witness environments make that objectification licensed?

The ULD materials state this as a universal invariant: every element can in principle be targeted, packaged, referenced, and re-entered as a “thing,” even if only metalinguistically and at high cost.

This predicts exactly the gradient we observe:

- contentful items (kinds, events) objectify cheaply in ordinary discourse;
- function words objectify expensively and usually require an explicit *mentioning practice* (teaching, editing, grammatical commentary).

RED therefore functions here as an **edge-probe** that maps the boundary between ordinary objectification and metalinguistic objectification, and it does so without abandoning the universality claim.

4.6.4 Interim conclusion

Case Study 3 is RED-weak by default—not because objectification is impossible, but because licensing is discourse-specialized and high-cost. This is not a defect; it is a principled prediction once cost becomes part of the explanatory vocabulary.

4.7 Quantitative and empirical footholds (computational and processing)

Although this chapter is primarily qualitative, the RED manuscript already sketches how the diagnostic logic can be used to supervise parsing and semantic role labeling—precisely because witness environments identify when event/property material should be treated as an argumental node.

Two empirical footholds are worth recording here, because they anticipate later chapters.

Computational pilot (NLP). A toy implementation reported in the RED manuscript claims improved subject detection for nominalized/clausal subjects relative to a baseline dependency parse (F1 0.84 vs. 0.76), with multilingual tests reported for English, Mandarin, Spanish, and Arabic.

In the book’s architecture, these results are not “applications added on”; they are *a second evidential channel* for the same hypothesis: if noun-slot stabilization is systematic, then pipelines that are trained or guided to respect witness environments should improve where ordinary parsers mis-handle promotion into argument positions.

Neurocognitive convergence (processing cost). The RED manuscript also summarizes a familiar convergence: object-like processing (often associated with temporal systems) and action-like processing (often recruiting motor systems) can co-activate when eventive material is nominalized, and unification demands recruit left IFG in phrase/sentence integration.

For the ULD framing, the key point is *not* to reduce grammar to anatomy, but to motivate a testable prediction: **higher-cost objectifications should show measurable processing signatures** (latency, acceptability, neural load) compared to low-cost ones—because the traversal requires additional packaging/unification work.

4.8 Alternative explanations and robustness checks

It is tempting to treat all of the above as “just Distributed Morphology” or “just Construction Grammar.” These traditions indeed get major pieces right: category behavior can be structurally conditioned (DM), and constructions can license stable interpretive outcomes (CG). But neither, by itself, explains why the same kind of witness success recurs so broadly that it motivates an architectural inference: that noun-slot stabilization is a privileged region of a universal hosting space rather than a scattered family of unrelated tricks.

3. CHAPTER 2 Background and Exi...

The ULD move is therefore not a rejection of existing work, but a change in explanatory target. Instead of asking which mechanisms *allow* nominalization, ULD asks what kind of *space of admissible positions* must exist for systematic objectification to be so reliable—and why languages differ primarily in **edge weights (costs)** rather than in the bare availability of objecthood.

4.9 Chapter summary

This chapter applied RED as an explicit witness procedure across three slices of reality and five languages.

Case Study 1 showed that kind/property material enters noun-slot behavior under low-cost genitive environments in ordinary discourse.

Case Study 2 showed that eventive material objectifies reliably, but that particular packaging routes (notably genitive-over-event hosts) can be moderately costly and construction-sensitive.

Case Study 3 showed that function words can be objectified primarily under metalinguistic conditions, yielding high-cost, RED-weak patterns that are nonetheless predicted once ULD is treated as an objecthood space.

The cross-case pattern is the central empirical moral: **the system does not collapse into “anything goes.”** Instead, it exhibits structured gradiePAnTs in cost and licensing—exactly what we should expect if RED is probing a universal domain whose internal geometry is stable, while languages differ in how they weight transitions among objecthood families.

Chapter 5. Extensions, Variants, and Edge Cases

5.1 Purpose and overview

The previous chapters introduced a claim and a method: **(i)** that languages can stabilize a striking range of material as **noun-slot entities**, and **(ii)** that this capacity is not a grab-bag of “special constructions,” but can be

tested by a disciplined diagnostic—**RED (Reverse Entailment Diagnostic)**—which reads nominal potential off *witness configurations* (possessives, nominalizers, determiner anchoring, quotation regimes, and related packaging). The purpose of this chapter is to do what articles rarely have space to do: to make the framework robust by confronting the cases where it could fail, where it becomes graded rather than binary, and where it must be extended to cover cross-domain and computational realities.

The guiding idea is simple. **A diagnostic is only as strong as the space in which it operates.** If RED tells us that an item can behave as a noun-slot entity, we also need a theory of *what “being a noun-slot entity” is*, what counts as *strong vs weak* evidence, and why some transitions are effortless while others feel marked. This is exactly the role of the Universal Linguistic Domain (ULD): not an optional add-on, but the **geometry of possible linguistic activity**—the constrained space of slots in which hosting, stabilization, and discourse addressability can occur at all.

Chapter 5 therefore does three things. First, it **extends** RED/ULD into a graded theory (cost/metric). Second, it presents **variants** of the framework (minimal, enriched, and computational). Third, it treats the **edge cases**—quotation-only nouns, function-word objectification, cross-media identity problems, and apparent counterexamples—and shows what would count as a genuine failure.

5.2 Internal extensions: from a binary diagnostic to a graded geometry

5.2.1 ULD = UNS and why “nounhood” stops being the discriminating variable

A central architectural refinement is the explicit axiom that **ULD is identical to a Universal Nounal Slot (UNS)**: the domain is not a neutral container where nouns occupy one region, but the domain itself is the space of linguistic “things” that can be made available for grammar and discourse as objects.

Once this axiom is adopted, the question is no longer “Can X become noun-like?”—because in principle everything is manipulable *as an object* at some cost. Instead, the question becomes: **How cheaply and robustly can X reach stable discourse objecthood under admissible packaging?**

This is the point where RED becomes more precise. In a noun-centered ULD, **RED is not “a path into nounhood.”** It is an **edge-probe**: it detects whether there is a licensed (low/moderate) path from an item’s current functional family into a hosting regime that yields robust noun-slot behavior (topic continuity, anaphora, quantification, modification, etc.).

5.2.2 Cost as the missing parameter

My ULD work makes the next step unavoidable: **introduce a metric.** A single theory must explain why *run* → *a run* feels ordinary while *and* → *the “and”* typically requires metalinguistic packaging, quotation, or explicit mention.

A practical definition treats **RED-Cost** as a composite markedness/availability measure estimated from multiple channels: (i) morphological/lexical marking, (ii) structural complexity, (iii) processing load, (iv) learnability/regularity, and (v) discourse stability (whether the result functions as a durable discourse object).

This yields a graded typology rather than a binary result:

- **RED-Strong**: cheap, productive, ordinary noun-slot behavior
- **RED-Mid**: available but restricted (constructional, register-based, semantically specialized)
- **RED-Weak**: primarily metalinguistic/quotational—interpretable but costly and marked
- **RED-Blocked**: no interpretable noun-slot behavior under admissible packaging (a genuine pressure point)

This preserves my original claim (“the architecture supports broad objectification”) while making it testable in the way a book demands: it predicts *where* costs accumulate and *how* languages differ—namely by weighting the same internal ULD edges differently.

5.2.3 Level structure as an extension of explanatory power

The same “graded geometry” idea extends naturally across levels of organization.

- **Level-0 (entry point): exemes**—modality-neutral expression units; pure form-patterns like *r-u-n* or [rʌn], prior to category and meaning.
- **Level-1 (primary (degree of) freedom): a three-region hosting model** (Ordinate/Subordinate/Coordinate), a one-term “tape” in which material can be stabilized as a simple term.
- **Level-2 (secondary (degree of) freedom):** a two-term predication relation ($t_1 \text{ R } t_2$) realized as the earliest syntactic derivational entity, a **predicate–argument skeleton (PAS)** (Dowty, 1991), with a special predicate connector and micro-slot structure, - the structural units of textual fabric.

(a) Level-2a: PAS as the germinal layer (nucleating matrix). At the most fundamental syntactic level—the first-order construction in the sense relevant to derivation—this two-term configuration yields a **predicate–argument skeleton (PAS)** as the earliest syntactic derivational entity: a bare but organized family of slots, an abstract argumentative frame that structures roles, slotting, and combinatorial possibilities prior to full sentence realization (Dowty, 1991; Halle & Marantz, 1993). In this sense, PAS functions as a **generative matrix** (a germinal / nucleating layer, a “reading frame”) produced when two one-term structures are bound through **predication**—through the predication relation RRR that converts mere hosting into a predicate–argument dependency.

(b) Level-2b: PAS-derived primary syntactic units (ready-to-go units). From PAS, the system produces modified forms of PAS—still Level-2 in kind, but higher *super-orders* of Level-2—namely the familiar primary syntactic elements that constitute the “ready-to-go” units of linguistic output: **phrases, clauses, sentences, and even certain compound words**. On this view, these units are not primitive. They are **second-order constructions**: elaborations of a first-order predicate–argument skeleton, produced by embedding,

- **Level-3 (tertiary (degree of) freedom):** supra-syntactical combinations (four-term structures) arising from coordination, scope, operator layering, and multi-unit composition. The functional units of textual fabric wrapping the higher-order field of discourse forces.
- **Level-4 (quaternary (degree of) freedom): text** as a higher-order field of discourse forces—topic management, rhetorical contrast, evaluation, genre constraints—operating over and through the lower structures without reducing to them but creating its own different shapes of rhetorical colors.

An “edge case” is often simply a case where we confuse levels: we treat Level-4 rhetorical objectification as if it were Level-2 grammatical nominalization, or we treat Level-0 exeme identity as if it were lemma identity. Chapter 5’s job is to prevent those confusions from becoming false counterexamples.

5.3 Variants of the framework

5.3.1 Minimal variant: RED as diagnostic, ULD as commitment

In the minimal variant, the book asks the reader to accept only two commitments:

1. **RED:** nominal potential is diagnosed by witness packaging and reverse entailment back to base-forms.
2. **ULD/UNS:** the diagnostic presupposes a space of noun-slot hosting where discourse objecthood is a legitimate grammatical outcome, not an anomaly.

This variant is sufficient to explain broad category flexibility, to compare languages by their available packaging strategies, and to expose the weakness of rigid lexical essentialism (e.g., strong TGG-style assumptions about intrinsically fixed category membership; Chomsky, 1970).

5.3.2 Enriched variant: exeme–lexeme–lemma layering and reverse entailment “upward”

A second variant becomes important once I integrate Lexical Ontology Persistence (LOP): we need to distinguish **what persists** from **how it is hosted**.

Here the exeme layer is crucial: the exeme is a modality-neutral expression pattern, independent of grammatical category, lexical status, or meaning; it is the lowest formal layer on which lexemes and lemmas are defined.

This enables a “reverse entailment ladder” in the other direction as well: not only do derived grammatical forms entail back to nominal potential (classic RED), but lexeme/lemma-level stabilization can entail back to the exeme as the persistent form-unit that makes cross-context identity talk coherent.

This enriched variant is where LOP sits most naturally: **LOP tracks persistence across contexts and media**, while **ULD tracks the geometry of hosting**, and **RED supplies the empirical bridge**—the test that tells us when a piece of material has entered a stable nounal hosting regime versus remaining a fleeting contextual profile.

5.3.3 Computational variant: ULD as a graph of placements; RED as pre-pass

A third variant treats the framework as computationally explicit.

On this view, ULD is not only a philosophical “space”; it is a **structured placement graph**. RED provides an “objectification detector” pre-pass: identify witness packaging (possessive/genitive, determiner anchoring, nominalizers, quotation), promote the packaged span to an argument-like node, and then allow parsing/SRL to assign roles to hosted nodes rather than raw tokens.

This directly targets a known pain point in NLP: nominalizations and property-as-object constructions disrupt naive predicate–argument extraction because material that looks verbal/adjectival is functioning as an argumental object.

The ULD placement view also reframes preposition ambiguity as a hosting problem: classify “to”-like forms by attachment/hosting regime (strong continuation vs weak framing vs fused infinitival slot exponent), rather than by surface word class.

5.4 Edge cases: where the theory is most likely to be misunderstood

5.4.1 Quotation-only and “mention” nouns

Some items appear noun-like only under explicit mention: *the “and”*, *that “the”*, *the “oh”*. These are not embarrassing exceptions; they are exactly what a cost-sensitive model predicts. They typically require quotation or metalinguistic packaging and therefore pattern as **RED-Weak** rather than RED-Strong.

The crucial test is not merely “can it appear with a determiner,” but whether it becomes a **stable discourse object**: can it be anaphorically tracked, modified, quantified, and integrated without repair readings. This is why discourse/accessibility cost is a decisive component of RED-Cost.

5.4.2 Function words and the boundary of objectification

Function-word objectification is the place where critics will say “this is trivial—you can talk about anything.” My framework’s reply is: **triviality disappears the moment we measure cost and stability**. We can allow

that “the and” is possible while still distinguishing it sharply from ordinary nominalization, and we can make that distinction empirically visible (frequency, register restriction, quotation dependence, anaphora behavior, processing signatures).

5.4.3 Strong vs weak prepositions and the hosting regime problem

A major family of edge cases arises from items that can be hosted in different places in the Level-2 geometry.

My corrected Level-2 architecture treats the Predicate Domain connector as **mirror-doubled**, with two coordinate ports, and a doubled Ordinate axis as the unique locus of verb realization.

This allows a principled split:

- **Strong prepositions:** predicate-integrated continuation markers hosted in the predicate connector’s object-facing coordinate port (profiling an unreachable target).
- **Weak prepositions:** object-domain framing heads hosted in the object domain’s coordinate region, forming PP-like coordinating objects (“under the table,” “on the chair”).

Computationally, this becomes a tractable placement classification problem; theoretically, it prevents prepositions from becoming a “miscellaneous” counterexample class.

5.4.4 Homophony of “to”: goal vs infinitival fusion

The hardest edge case is where *the same surface string* belongs to different hosting regimes. My ULD work treats “to” as homophonous: goal/prepositional “to” as predicate-side continuation vs infinitival “to” fused into infinitival slot exponents.

This is exactly the kind of case where traditional category labels break down, and where ULD-style placement resolves the confusion without multiplying lexical entries unnecessarily: what differs is not “word class,” but **hosting location and structural role**.

5.5 Cross-linguistic and cross-domain extensions

The article’s multilingual core (English, Mandarin, Tongan, Spanish, Arabic) already shows that nominalization pathways differ in surface machinery while supporting the same functional outcome: entity stabilization in noun-slot positions (e.g., possessive/genitive strategies). This chapter adds a stronger claim: **typology belongs inside the metric**.

Languages do not differ primarily in whether they allow objectification (they all do); they differ in **which edges are low-cost**: whether nominalizers are obligatory, whether determiner shells are required, whether quotation is the only viable route for certain items, and how productive each transition is.

Cross-domain extensions follow the same logic. Scientific language, pedagogy, legal language, and programming discourse all increase the rate of high-cost objectification because they require reference to the machinery of representation itself (operators, constructions, frames). In ULD terms, these are contexts where the discourse field (Level-4) systematically biases which nounal families are foregrounded and which hosting transitions are repeatedly exercised.

5.6 Limits and boundaries: what would count as failure

A serious framework must state its falsifiability conditions.

In my cost-sensitive RED typology, a genuinely problematic case is not “X is marked,” but **RED-Blocked**: an item for which no admissible packaging yields interpretable noun-slot behavior, even in metalinguistic regimes, and where this failure is stable across speakers/contexts.

Even then, the interpretation would not automatically be “RED is false.” It could instead indicate:

- a missing hosting mechanism in the ULD map (an unmodeled slot or connector),
- a mis-specified notion of witness packaging, or
- a level-confusion (mistaking discourse quotation phenomena for grammatical nominalization, or vice versa).

This is why Chapter 5 treats “limits” as theoretically productive: they tell us where the geometry needs refinement.

5.7 Alternative formulations and design choices

This book commits to the claim that ULD is a constrained geometry, but several design choices remain open (and are healthy to keep open at the monograph stage):

1. **Discrete vs continuous metrics**: do we model RED-Cost as a scalar, or as a vector across its components (marking vs processing vs discourse stability)?
2. **Family inventory**: how many nounal families are needed beyond the minimal set (N-Ref, N-Pred, N-Op, N-Link, N-Pack), and what invariants best define them?
3. **Graph-theoretic implementation**: should ULD be implemented as a directed graph with weighted edges, or as a typed feature-structure space with costs derived from description length (MDL-style) regularities?
4. **Diagnostics beyond possession**: possession is powerful (Partee, 1997), but witness packaging must also include determiners, nominalizers, quotation regimes, and clausal shells; each widens coverage but increases the risk of overgeneration.

These alternatives matter because they control what the theory predicts about gradience, typology, and computational learnability.

5.8 Feedback to the core theory: what the edge cases teach us

The net result of the extensions and edge cases is not to weaken the core claim, but to sharpen it:

- The central phenomenon is not “category chaos,” but **context-sensitive stabilization in a constrained space of possible hostings**.
- The correct explanatory target is not “does X become a noun,” but “what is the **licensed path** (and its cost profile) by which X becomes a **stable discourse object**.”
- The strongest unification comes from treating **ULD as placement geometry** and **RED as an evidential probe** operating inside that geometry, rather than treating RED as a standalone trick.

This feedback loop also clarifies how exemes belong in the story. The more we push on cross-media and cross-level identity, the more we need a lowest formal layer that can persist while higher layers (lexeme, lemma, constructional profile) vary. That is exactly what the exeme provides: a modality-neutral expression unit—pure pattern—on which lexemes and lemmas are built.

5.9 Chapter summary and bridge to Chapter 6

This chapter extended the book’s core proposal in three directions.

First, it moved from a binary to a **graded** account by introducing **RED-Cost** and a strength typology (Strong/Mid/Weak/Blocked), where the decisive issue is robust discourse objecthood rather than mere structural occupancy.

Second, it formalized multiple **variants** of the framework: a minimal diagnostic variant, an enriched exeme–lexeme–lemma variant compatible with lexical persistence concerns, and a computational variant in which ULD becomes a placement graph and RED becomes an objectification detector that repairs parsing and role labeling.

Third, it confronted **edge cases**—quotation-only nounhood, function-word objectification, preposition regime ambiguity, and “to” homophony—showing that these are not exceptions to be hidden, but stress-tests that reveal where the geometry must be explicit (especially at Level-2).

Chapter 6 will take the next step: it will treat these extensions not only as theoretical refinements, but as **testable predictions** in neurolinguistics and computational modeling—where “cost,” “hosting,” and “stabilization” can be operationalized via processing signatures and performance on structured NLP tasks.

Chapter 6 — Connections to Neuroscience, Computation, and Practice

6.1 Purpose and Overview

Up to this point, the book has built an internal architecture: (i) a diagnosis of noun-slot behavior via Reverse Entailment (RED), (ii) an explicit hosting geometry (ULD/UNS) that makes “slotting” a formal object rather than a loose metaphor, and (iii) a layered account of lexical identity (exeme → lexeme → lemma) sufficient to talk about persistence across contexts and realizations. Chapter 5, in particular, presents RED in full and shows how it probes the noun-slot region of ULD, with implications that naturally extend toward implementable parsing and semantic-role criteria.

Chapter 6 asks a different kind of question: if the internal picture is even approximately correct, what should we expect to see (a) in minds and brains, (b) in computational models that actually have to make predictions token-by-token, and (c) in applied work that forces representational commitments (annotation, lexicography, pedagogy, clinical description)? This chapter is not a general review of neurolinguistics or NLP. It is a conceptual mapping from the book’s objects and constraints to empirical and engineering traditions that can test them.

The thread tying the chapter together is a single contrast: stable structure versus context-driven activation. The core claim of ULD/UNS is not that “context matters” (everyone grants that), but that context matters through a constrained set of admissible positions—slots—within which linguistic material can be stabilized and made inferentially available. Neuroscience and computation are useful here because they force two disciplines to make explicit what linguistics often leaves implicit: what persists, what changes, and what the cost of a transition should be.

6.2 Cognitive and Neural Foundations

6.2.1 How word representation is commonly modeled

Psycholinguistic work has long treated “the mental lexicon” as a structured memory system supporting rapid access, morphological decomposition, and meaning construction (Levelt, Roelofs, & Meyer, 1999). Contemporary cognitive neuroscience tends to reject the idea of a single localized “word cell” in favor of distributed representations: words (and word-like units) correspond to patterns of activity and connectivity

spanning temporal, frontal, and parietal regions, with modality-specific contributions for speech, reading, and action-related semantics (Pulvermüller, 2005).

Two empirical phenomena matter especially for the present framework:

1. **Time-sensitive lexical access and composition.** Event-related potential studies (e.g., N400 effects) show that lexical-semantic expectations modulate processing rapidly, implying that contextual fit can gate which parts of lexical knowledge are activated (Kutas & Hillyard, 1980; Lau, Phillips, & Poeppel, 2008).
2. **Unification and slotting operations.** Neurocognitive models that distinguish stored lexical knowledge from combinatorial operations (often linked to left inferior frontal regions) align naturally with the book's separation between persistent lexical structure and constructional hosting (Hagoort, Hald, Bastiaansen, & Petersson, 2004; Hagoort, 2013).

The empirical moral is not that the brain “contains ULD.” It is that a division between (i) relatively stable representational structure and (ii) context-sensitive selection/unification is a well-motivated organizing principle for interpreting linguistic behavior in neural terms.

6.2.2 Interpreting RED and ULD in neural terms

The book's internal architecture invites a straightforward compatibility claim: persistent lexical structure corresponds, at a coarse grain, to relatively stable connectivity patterns, while context supplies activation and selection over that structure. This is precisely the kind of mapping Chapter 6 is designed to articulate: lexical ontology as long-term structure; contextual “slices” as time-indexed activation.

To make that mapping precise enough to be testable—without pretending to be a neural theory—we can align the book's main objects with three neurocomputational roles:

(a) Exeme-level persistence as long-term representational scaffolding.

An exeme, in my system, is the identity-bearing unit across realizations and contexts; a lexeme is its language-specific grammatical projection; a lemma is the citation label used by reference systems.

In neural terms, this predicts that across surface variation (e.g., inflectional forms, phonological variants, orthographic variants), there should be a recurring representational core—realized as partially shared activation patterns or shared representational geometry—while still allowing graded divergence as meaning, register, and constructional role vary.

(b) ULD/UNS as the space of admissible “positions” for stabilization.

ULD is not “another module,” but a constraint space: a geometry of slots in which linguistic material can be stabilized as entity-like, predication-like, or operator-like material (as developed in my multi-level description of term assembly and term composition). One direct computational reinterpretation of this is that “POS tagging” is better framed as *positional hosting inference* over latent micro-states rather than assignment to universal classes.

The cognitive analogue is that the brain's combinatorial system may track *role-like* states (subject-like, predicate-like, determiner-like, frame-like) that are not identical to traditional part-of-speech labels.

(c) RED as a controlled transition test, not a free coercion license.

RED's theoretical point is disciplined: it treats nominalization potential as something that can be *witnessed* by specific derivational environments (possessives/genitives, nominalizers, metalinguistic mention, etc.), and then reverse-entailed back to base material. In cognitive terms, RED predicts that “entity stabilization” should carry measurable signatures: increased unification load when an item is forced into a noun-slot against its default profile, and facilitation when prior context has already prepared that slot.

This is the level at which the framework becomes empirically risky (and therefore valuable): it does not merely predict that context will affect processing, but that *specific* context types—those that instantiate noun-slot hosting—should systematically modulate the costs and the representational overlap patterns observed in priming, ERP, and neuroimaging.

6.2.3 Testable predictions and constraints

The chapter template emphasizes that the theory gains credibility when it generates concrete predictions.

Here are four that follow directly from RED+ULD as presented in the book:

1. **Cross-context “same item” recurrence with structured divergence.**

If exeme-identity is real (not merely a lexicographic convenience), then across contexts that vary in argument structure and discourse function, one should observe a stable representational backbone with systematic modulations tied to slotting. Representational similarity analysis (RSA) over MEG/EEG-informed source estimates or fMRI patterns is a plausible tool for testing this (cf. Huth et al., 2016 for semantic mapping logic; methodologically).

2. **Nominalization as a predictable increase in composition cost when it is “forced.”**

Sentences that require a verb-like or function-word-like item to occupy noun-slot positions should elicit measurable cost relative to matched controls—especially when the construction is marginal and relies on metalinguistic licensing. This is compatible with unification-centered accounts of frontal involvement (Hagoort, 2013) and with findings that nominalized verbs recruit both action-related and object-related circuits (Pulvermüller, 2005).

3. **A measurable “cost of hosting” that aligns with expectation/surprisal.**

My ULD discussions already motivate a metric notion of structural cost that can be operationalized via surprisal-style processing models: unlikely placements (e.g., a determiner hosted where a qualifier is expected) predict increased processing load and acceptability degradation.

This is directly testable with reading-time and ERP paradigms, and conceptually aligned with expectation-based parsing (Hale, 2001; Levy, 2008).

4. **Lesion/dissociation predictions: persistence versus placement.**

If we separate persistent lexical structure (exeme/lexeme) from slotting operations (ULD-mediated stabilization), then aphasia and related disorders should sometimes show partial dissociations: preserved lexical knowledge with impaired role assignment/unification, or preserved syntactic scaffolding with degraded lexical specificity. This matches the general logic of dissociation reasoning in cognitive neuropsychology (Caramazza, 1986) and gives a principled way to interpret “impaired lexical persistence” patterns as the template suggests.

6.3 Computational Modeling and NLP

6.3.1 What current models do—and what they systematically miss

Modern NLP systems are exceptionally good at capturing distributional regularities: embeddings learn graded similarity from co-occurrence (Mikolov et al., 2013), and transformer models learn contextual representations that shift with local and global prompt conditions (Devlin et al., 2019). These models therefore align with the “fluidity” side of the book’s motivating contrast: they instantiate context sensitivity at scale.

But the same success exposes a limitation that the chapter template explicitly asks me to foreground: systems optimized for next-token prediction may model meaning largely as *contextual activation* while underspecifying *persistent lexical structure*—especially historical layering, cross-media identity, and principled transitions into noun-slot behavior.

In practical terms, many models behave as if “identity” is an emergent statistical convenience rather than an object with constraints.

My framework is not anti-transformer; it is a proposal for an additional representational commitment: a persistent lexical store (vertical) plus a context-driven selection mechanism (horizontal).

6.3.2 A ULD-compatible model sketch: persistent store + positional hosting

A computational implementation faithful to the book’s architecture can be described at three interacting layers:

Layer 1: A persistent lexical store indexed by exeme identity.

Represent each exeme as a structured object (e.g., a graph or typed bundle) encoding: (i) a conceptual core, (ii) sub-sense hierarchy, (iii) historical strata when available, and (iv) cross-modal exemplars. Lexemes are then language-specific projections that add morphosyntactic paradigms and subcategorization information; lemmas are merely labels for lookup and reference.

Layer 2: A ULD slot-geometry decoder (“term assembly”).

Treat the first pass over text as assignment of tokens/exemes to ULD term regions and micro-positions. This reframes “POS tagging” as positional hosting inference with latent micro-states rather than class labeling.

Computationally, this can be implemented as:

- an HMM/CRF sequence model with adjacency constraints reflecting the term tape, or
- a transformer with an auxiliary “term-assembly head” that predicts positions and boundaries and feeds them into downstream parsing.

Layer 3: RED as an evidential operator over candidate structures.

Once a candidate structure is produced, RED becomes a checkable constraint: given a suspected noun-slot environment, test whether the construction licenses entity stabilization for the item and whether reverse entailment supports nominal potential. Operationally, this can be implemented as a combination of rule-templates (for overt nominalizers/possessives) and learned classifiers (for less overt cases like clausal subjects or metalinguistic mention).

This yields a division of labor that ordinary pipelines blur: transformers provide flexible contextual representations; ULD provides an explicit positional geometry; RED provides disciplined diagnostics for entity stabilization.

6.3.3 Experiments that make the framework computationally falsifiable

The template recommends concrete comparisons between standard models and ontology-enriched models.

Here are experiments that align tightly with my claims:

1. Nominalization and noun-slot detection as structured prediction.

Benchmark on constructions where noun-slot behavior is the central difficulty: clausal subjects, gerunds, possessive attachment, metalinguistic mention, and “function-word as discussable item” contexts. Evaluate whether explicit ULD positions improve F1 over POS-only features on UD parsing and SRL tasks (Nivre et al., 2016; Palmer, Gildea, & Kingsbury, 2005).

2. Unsupervised induction: from POS tags to UNS positions.

Train latent-state models (HMM/CRF/transformer taggers) where states correspond to UNS micro-positions rather than POS classes, with constraints encoding adjacency and allowable transitions. Test whether induced states generalize better cross-linguistically than traditional tagsets—especially in low-resource settings.

3. Hosting cost as surprisal and as human difficulty.

Define a “hosting cost” for assigning a token to a micro-position and compare it to human reading times and ERP indices. The prediction is not merely “higher surprisal, harder processing,” but “higher surprisal specifically when slotting violates ULD adjacency/compatibility.”

4. **Cross-media consistency tasks.**

If exeme identity is genuinely cross-modal, evaluate whether models enriched with persistent lexical objects show improved alignment between speech and writing representations (e.g., ASR transcripts vs. written corpora) and improved robustness to orthographic variation. This operationalizes the book's insistence that identity is not exhausted by one medium.

6.3.4 Direct engineering payoffs

My book already identifies applied tasks that should benefit directly from term-assembly representations: information extraction/entity linking (better nucleus vs. qualifier identification), semantic parsing/slot filling (prepositional framing as early relational cue), and controlled generation (enforcing well-formed assembly to reduce agreement/modifier errors).

The broader claim is that these are not merely “applications” appended to a theory. They are stress-tests: if explicit positional structure improves downstream behavior, the theory is not just descriptively adequate; it is computationally useful.

6.4 Practical and Applied Implications

6.4.1 Implications for NLP systems and annotation practice

NLP annotation schemes often hard-code assumptions about categories: POS inventories, dependency relations, and sense inventories. My framework suggests a different priority ordering:

1. annotate *positions* (ULD micro-positions and term boundaries) as primary latent structure;
2. treat category labels as secondary summaries of positional regimes;
3. use RED-like diagnostics to decide when a token should be treated as entity-like for downstream tasks (coreference, entity linking, SRL).

This is a concrete proposal for harmonizing cross-linguistic annotation where POS systems misalign: the shared object is the slot-geometry, not the label set.

6.4.2 Education and literacy

A persistence-oriented lexical view implies pedagogical changes: teach words not as flat definitions, but as structured semantic organisms with layered extensions and cross-context behavior. This aligns with what many educators do informally (roots, families, metaphors) but gives a principled rationale: learners are not memorizing lists; they are building stable lexical objects that can be activated and re-profiled under constructions.

6.4.3 Clinical and diagnostic relevance (conceptual, not clinical)

Even without proposing clinical protocols, the framework provides a language for describing breakdown patterns: failures of slotting/unification versus degradation of lexical cores; partial preservation of older layers versus newer extensions; or impaired transitions into noun-slot stabilization. This is precisely the sort of “conceptual applicability” the chapter template recommends, without overclaiming clinical authority.

6.4.4 Lexicography, corpus work, and language documentation

Conventional dictionary entries privilege lemma-headwords and flatten sense structure into lists. My framework suggests richer entries: vertical timelines (diachronic layering), cross-media exemplars, and explicit representation of a persistent ontological core rather than only distributional sense partitions.

Corpus annotation can likewise be redesigned to store not merely tokens and tags, but token-to-exeme link hypotheses and slotting histories across discourse.

6.5 Triangulating Theory, Brain, and Machine

This chapter's strongest methodological claim is triangulation: theory proposes structure, brain data provide constraints and plausibility, and computational models provide explicit implementations whose failures are informative.

Convergence across the three strengthens the book's central commitments. For example, if (i) ULD-style positional heads improve parsing and SRL, (ii) surprisal-like hosting costs correlate with human difficulty, and (iii) neurocognitive measures show recurring representational backbones with slot-driven modulations, then "universal noun-slot hosting" ceases to be a clever descriptive generalization and becomes a constrained research program.

Equally important, divergence is productive. If a model works but neural evidence is unclear, that suggests the model may be exploiting engineering shortcuts rather than capturing the intended cognitive decomposition. If neural evidence supports stability/activation distinctions but slot-geometry constraints do not improve computation, that suggests either the wrong formalization of slots or the need for richer training signals (diachronic and cross-modal data). Either way, the framework is set up to be revised by external pressure rather than insulated from it.

6.6 Chapter Summary

Chapter 6 has argued that the book's internal architecture naturally extends outward: RED and ULD make specific claims about what persists and what changes, and these claims can be probed by psycholinguistic measures, neurocognitive constraints, and computational performance. The key idea is stable lexical structure plus context-sensitive activation, with noun-slot stabilization as a privileged site where the transition becomes visible and measurable.

The most direct empirical predictions are (i) recurring representational backbones for the "same" lexical object across contexts, (ii) cost signatures for forced noun-slot stabilization, and (iii) measurable hosting costs that align with expectation/surprisal when ULD constraints are violated. On the computational side, the agenda is explicit: implement a two-layer system (persistent store + activation/selection), integrate ULD positional inference as an inductive bias, and use RED as a diagnostic operator over candidate structures.

In the next chapter, we return from these external connections to the internal methodological and philosophical implications of taking ULD/UNS seriously as a constrained geometry of linguistic activity—especially the consequences for what counts as an explanation, what counts as evidence, and how category talk should be reconstructed once "position" becomes the primary formal object.

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Chapter 7

Applications and Examples in Real Life and Research

7.1 From a diagnostic to a usable instrument

By this point in the book, RED has done its primary job: it provides an *empirical* way to test whether an item can be stabilized as a noun-like entity (i.e., hosted in noun-slot positions) under a witness construction, and it lets us reverse-infer “nominal potential” from that stabilization. What matters for applications is that RED is not merely a clever test; it is a *repeatable procedure* that yields a structured output: **(i)** a candidate span, **(ii)** a witness regime (possessive/genitive/nominalizer/quotation, etc.), **(iii)** a host position, and **(iv)** a graded sense of “how costly” or “how natural” the stabilization is in context.

The Universal Linguistic Domain (ULD), in turn, is what makes that output interpretable. RED tells us *that* an item can be hosted as an entity; ULD tells us *what it means*—structurally—when it is hosted, because ULD is the constrained space of possible discourse-positions within which hosting, profiling, and stabilization occur at all.

Put bluntly: **RED is the probe; ULD is the geometry the probe is probing.**

That pairing is what allows Chapter 7 to be more than “possible implications.” It allows a shift from *theoretical description* to *operational use*: in linguistic research design, in annotation practice, in NLP pipelines, and in real-world professional contexts where category flexibility is not a philosophical curiosity but a practical source of error, ambiguity, or misunderstanding.

7.2 Research applications in linguistics and semantics

One immediate payoff of the RED→ULD transition is that “category flexibility” stops being treated as an unstructured list of constructions and becomes something we can *map*. In ULD terms, a token’s category is not an intrinsic label but a placement outcome (the positional category principle).

This reframes several classic research programs.

First: typology and cross-linguistic comparison.

Instead of asking whether a language “has nominalization,” we can ask which *witness regimes* are productive, which *hosted noun-slot outcomes* they license, and how often stabilization requires extra scaffolding (overt nominalizers, determiner anchoring, genitives, quotatives, etc.). RED naturally decomposes “nominalization” into comparable signals across typologically diverse systems (cf. the cross-linguistic marker inventories in my RED dataset).

Second: constructional semantics and coercion.

Many debates about coercion can be rephrased as disputes about *where* coercion lives: in the lexicon, in the construction, or in discourse. ULD gives a clean structural hypothesis: much of what is called “coercion” is really **host-driven stabilization**—a construction forcing an item into an entity-profile region of the domain. RED provides a way to test when such stabilization is truly occurring (witnessed and reversible) versus when an analyst is merely re-describing the effect.

Third: discourse and metalinguistic regimes.

Quotation, mention, and “talking about words” often look like special cases. Under RED+ULD, they are not special in kind; they are witness regimes that reliably package material as an entity so it can participate in discourse management (topic tracking, contrast, evaluation). This is exactly the kind of register-shift condition where RED predicts measurable changes in objectification frequency and where computational systems often fail.

A useful way to make this concrete is to treat RED outcomes as a *layer* we can annotate.

A minimal ULD–RED annotation layer (illustrative):

Field	What it records	Why it matters
span	the candidate packaged unit (token or multiword)	RED applies to spans, not only single words
witness	possessive/genitive/nominalizer/quotation/etc.	it defines evidence type and comparability
host_slot	which noun-slot is satisfied (subject/object/NP-internal)	ties directly to UNS claim
ULD_level	whether stabilization is local (term-level) or PAS-level	separates “token trick” from structural consequence
RED_cost	low/medium/high (or scalar)	links to acceptability/processing predictions

This is “book architecture” turning into “research architecture”: the same analysis becomes portable across corpora, languages, and experimental settings.

7.3 Cognitive and psycholinguistic applications

The RED→ULD transition also clarifies what to measure in cognition. If the universal noun slot is a real cognitive capacity, then stabilizing an item as an entity should show *signatures*—in time, in error patterns, and potentially in neural indices—especially when stabilization is costly or marginal.

A simple working hypothesis is:

- **Low-cost hosting** (highly conventional nominalizations, well-entrenched conversions) should behave like ordinary noun-processing in comprehension and production.
- **High-cost hosting** (rare objectifications, forced mention-like packaging, unusual function-word nominalization) should recruit extra processing resources and show sensitivity to context and task.

This aligns naturally with well-studied psycholinguistic contrasts between lexical access, integration, and reanalysis (e.g., N400-like semantic access vs. P600-like reanalysis effects; Kutas & Hillyard 1980; Friederici 2011). It also connects to “unification” style accounts of sentence-level integration (Hagoort 2005), where the core prediction is not “a noun module exists,” but that **entity-stabilization is an integration operation whose difficulty scales with RED-cost**.

Two experimentally clean paradigms follow almost immediately:

1. **Acceptability × processing gradient alignment.**
Collect graded acceptability judgments for a family of RED candidates (varying witness regime and span length), then test whether reading times / ERP amplitudes covary with the same gradient. ULD’s contribution here is that we are not correlating arbitrary “weirdness” with time; we are correlating *position-forcing* in a constrained slot space.
2. **Task-dependent regime shifts (ordinary discourse vs. explicit mention).**
Move participants between tasks that encourage “talking about words” (metalinguistic instruction, quoting, spelling, definitions) and tasks that encourage ordinary reference. RED predicts that the *rate and type* of objectification changes systematically, not randomly, because the discourse task changes which slots must be satisfied.

Neuroscientifically, this sits comfortably beside distributed lexical-semantic activation accounts and combinatorial/integration accounts (Pulvermüller 2005; Hagoort et al. 2004), while adding a sharper claim about **slot-level stabilization** as the operative unit of explanation.

7.4 NLP and AI: making “objectification” a first-class operation

In computational linguistics, the practical pain point is familiar: parsers and SRL systems often mis-handle nominalizations, proposition-as-argument structures, and packaging constructions, precisely because surface POS labels do not tell you when the grammar has turned something into an argument-domain.

My RED computational framing is explicit about the remedy: detect witness packaging, promote the packaged span to an argument-like node, and then run role assignment on the updated structure.

This redefines “nominalization handling” as a **graph repair problem**, not as a bag of exceptions.

7.4.1 Nominalization-aware parsing and SRL

A RED-informed pipeline can be described as a modular stack:

1. language-specific witness inventory
2. candidate span extraction (including multiword)
3. witness validation (grammatical/interpretable check)
4. parse update: promote to hosted nodes
5. SRL on hosted nodes

That design directly targets event nominalizations and property-as-object constructions, where “things that look verbal/adjectival” must be treated as arguments.

And it produces falsifiable evaluation targets: if RED is doing real work, you should see improvements exactly where objectification matters (nominalization-aware attachments; proposition-as-argument role labeling; robustness under quotation/mention spikes).

My article goes as far as providing pseudocode and explicit evaluation expectations (targets and baseline deltas) for a RED-enhanced nominalization parser.

7.4.2 Placement regimes: “to” and prepositions as an applied case study

ULD becomes especially useful when the same surface form belongs to different hosting regimes. The “to” system is the clearest example in my framework: a goal/prepositional “to” as a continuation connector versus an infinitival “to” fused into slot exponents (e.g., “to be”), i.e., not a free connector at all.

Computationally, that is not “POS tagging”; it is a placement classification problem whose features are structural (attachment site, scope, PP-like object formation, fusion behavior).

A RED-informed model treats these outcomes as hosting decisions—so it should generalize better across lexical items and in low-resource settings where surface heuristics collapse.

7.5 Lexicography, annotation, and the exeme/lexeme/lemma interface

Many practical disagreements in lexicography and corpus work are really disagreements about *what the unit is*. A dictionary wants stable entries; corpora want stable indices; speakers produce fluid uses. My architecture gives a disciplined compromise:

- **Exeme:** the modality-neutral, minimal expression unit that can persist across media and contextual shifts (the “same” item across speech/writing/orthographic variation).
- **Lexeme:** the language-specific lexical organization that clusters conventionalized patterns, morphological families, and entrenched usage.
- **Lemma:** the indexing convention imposed by tools and practices (dictionary headwords, annotation IDs, normalization rules).

RED then provides evidence about how far a given exeme/lexeme can be hosted as an entity, and ULD provides the slot-geometry in which that hosting is described.

In practice, this suggests a lexicon that stores not only “senses,” but **hosting profiles**:

- typical witness regimes (what packages it)
- typical hosted positions (where it lands)
- a conventionality measure (RED-cost distribution)
- cross-media attestations (where it appears as “the same thing”)

This is exactly the kind of structure that can bridge traditional resources (WordNet; FrameNet; PropBank) to modern contextual models: lexical resources become *constraints and affordances* on placement, not merely lists of meanings (Fellbaum 1998; Baker et al. 1998; Palmer et al. 2005; Devlin et al. 2019).

7.6 Education, editing, translation: practical clarity about “what counts as a thing”

In classrooms and professional writing, “parts of speech” are often taught as if they were intrinsic labels, and then students are punished for the very flexibility that real language uses constantly. A RED-first pedagogy fixes this without collapsing into “anything goes.”

A simple teaching routine mirrors the diagnostic:

1. Identify a base expression in context.
2. Apply a witness packaging strategy (possessive, nominalizer, quotation).
3. Check whether it behaves like an entity (subject/object/NP-internal).
4. Compare low-cost vs. high-cost cases, and ask what the construction is doing.

This makes category fluidity concrete and testable, while keeping the category system intact: categories are *stabilized outcomes* of hosting, not discarded labels.

In translation and interpretation, the same clarity matters because languages differ in preferred witness regimes. RED’s cross-linguistic inventory approach supports a practical workflow: instead of translating “nominalizations” as a class, the translator tracks **what is being turned into an entity** (event, property, relation, function word) and chooses the target-language packaging strategy that produces the same hosted discourse effect.

7.7 Across “slices of reality”: media, modality, and discourse environments

A core motivation behind introducing exemes and a domain-geometry is that words do not live in one medium. The “same” linguistic material appears in speech, writing, typography, hashtags, UI buttons, memes, legal headings, classroom diagrams, and citation practices—each of which invites different stabilization strategies and different discourse slots.

ULD provides a way to say that this is not merely sociological variation. It is a structured interaction between:

- the available slots and chains in the domain (what can be foregrounded, packaged, or tracked), and
- the medium’s affordances (what forms of packaging are cheap, salient, or conventional).

This is also where “quantum-style” intuitions can be used carefully: interpretation is often context-selected, but what my framework adds is a *constrained slot space* and a method (RED) for determining when selection has produced genuine entity-stabilization rather than mere ambiguity talk.

7.8 Limits, risks, and what not to claim

Application-minded frameworks are easy to overextend. Three cautions keep the project honest.

1. **RED is a diagnostic, not a metaphysical proof.** It supports an inference about nominal potential under witness regimes; it does not by itself settle what “meaning” is.
2. **ULD is a constraint space, not a replacement for sociolinguistics.** It can describe where discourse forces operate, but it does not replace explanations that live in institutions, power, genre, or community practice.

3. **Computational success must be benchmarked.** Proposed gains are meaningful only when tied to explicit datasets, model families, and error analyses—especially for the hardest cases (quotation spikes, marginal objectifications, and multiword spans).

7.9 Chapter summary

This chapter has treated RED and ULD as tools we can *use* rather than ideas we can merely endorse. RED becomes actionable when its outputs are treated as hosted-node decisions (with costs and evidence types), and ULD makes those outputs structurally meaningful by placing them inside a constrained geometry of possible discourse positions.

In research, this pairing turns category flexibility into a mappable phenomenon: witness regimes, hosted positions, and gradients. In cognition, it yields testable predictions about processing cost and regime shifts. In computation, it suggests concrete pipeline repairs for parsing and SRL—promoting packaged spans to argument nodes—and reframes difficult items (like “to”) as placement problems.

In practice—education, lexicography, translation—it provides a disciplined way to answer a surprisingly common question: **when does language treat something as a “thing,” and how does it do so reliably?**

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Chapter 8

Critiques, Objections, and Replies

8.1 Why this chapter exists

Any framework that makes a universality claim will attract two legitimate reactions: interest and resistance. Resistance is not noise; it is how a theory is clarified. This chapter therefore treats criticism as part of the method. The preceding chapters advanced three linked commitments: **RED** as a disciplined diagnostic for noun-slot stabilization, **ULD/UNS** as a formal geometry of admissible discourse positions that makes that stabilization structurally intelligible, and the **exeme** → **lexeme** → **lemma** layering that separates persistent form-units from language-specific lexical organization and from indexing conventions. The present chapter stages the most predictable objections and replies to them in a form that mimics the logic of peer review: each objection is stated in its strongest plausible version, then answered by clarifying what the framework claims, what it does not claim, and what would count as decisive evidence against it.

8.2 Objection family I: “This is trivial—of course we can talk about anything”

Objection.

The ability to treat almost anything as a noun-like “thing” is trivial. Humans can mention words, quote them, and discuss function words explicitly. That does not imply a universal noun slot as a structural feature of grammar; it only reflects a general metalinguistic capacity.

Reply.

The objection is correct about one part of the phenomenon and incorrect about the target of explanation. The framework does not deny metalinguistic mention; it uses it as a controlled limiting case. The claim is not that “mention exists,” but that grammars reliably support **stabilization** of diverse material as discourse objects under identifiable witness regimes, and that this stabilization is structured rather than flat—exactly the kind of structure that older category-essentialist views tend to treat as marginal or exceptional (Chomsky 1970), and that later “dynamic” morphosyntactic and constructional views treat as systematic but often under-constrained (Halle & Marantz 1993; Goldberg 1995).

Two clarifications separate nontrivial structure from trivial availability:

1. **Cost and robustness, not bare possibility.**

If the theory were “anything can be talked about,” it would predict no internal organization. But the framework predicts systematic gradients: some objectifications are low-cost and ordinary (e.g., productive event nominalizations; entrenched conversions), while others are high-cost and restricted to explicit mention/quotation regimes. This is exactly the empirical gap between “possible” and “stable” that a diagnostic is meant to measure, and it aligns with long-standing observations that possession/genitives are a particularly strong test bed for noun-like behavior (Partee 1997).

2. **Positionally licensed outcomes, not free creativity.**

ULD is introduced to explain why stabilization is possible at all: it is the constrained space of admissible discourse positions. If speakers can stabilize an interjection, a preposition, or a clause as an entity, the claim is that this is a **hosting outcome**—a structural placement result—rather than a mere conversational “anything goes” principle. This logic is compatible with constructional accounts of category emergence (Goldberg 1995) and with root-based models in which categorization is an outcome of functional structure (Halle & Marantz 1993), but it adds an explicit *domain-level* constraint system: slot-geometry, reachability, and cost.

8.3 Objection family II: “RED confuses morphology with semantics”

Objection.

RED relies heavily on derivational witnesses (possessives, nominalizers, genitives). But morphological availability does not entail semantic entityhood. A language can permit a certain form without implying that the base has genuine nominal potential.

Reply.

This is a fair worry, and it is exactly why RED is defined as a *multi-step procedure* rather than a one-step “attach -’s and declare victory.” The evidential point is not “a suffix exists,” but that a resulting structure behaves like a discourse object and can occupy noun-slot positions under controlled witness regimes—precisely the kind of interface test that motivates the original comparison to semantic entailment reasoning (Dowty 1991) and to genitive-based diagnostics (Partee 1997).

Two further constraints tighten the reply:

- **Interpretability and discourse stability.** A witness form that is morphologically well-formed but semantically unusable (or usable only under repair) is not treated as RED-strong evidence.
- **Host-slot verification.** The relevant test is noun-slot occupancy (subject/object/NP-internal), not morphological derivation alone—an approach consistent with the broader move away from “intrinsic” categories (Chomsky 1970) toward structural or constructional licensing (Halle & Marantz 1993; Goldberg 1995).

This reply also yields a falsifiability condition: if a language provides packaging morphology but the packaged result cannot function as a stable noun-slot object (cannot be tracked, modified, quantified, or integrated without systematic repair), then that witness is not accepted as evidence for low-cost nominal stabilization.

8.4 Objection family III: “Your examples collapse into ‘coercion’ or ‘type-shifting’”

Objection.

Formal semantics already allows type-shifting and coercion: predicates can be shifted to individuals, events to objects, properties to kinds. Your noun-slot effects are just instances of type-shifting, not evidence for a new architecture.

Reply.

Type-shifting accounts are compatible with many descriptive facts, but they often leave two issues underdetermined:

1. Why the shifts are so constructionally reliable and gradient.

If coercion is treated as a freely available repair, it becomes difficult to explain the patterned differences between low-cost, productive objectification and high-cost, mention-like objectification. The ULD move explicitly targets this: it reinterprets coercion as **host-driven stabilization in a constrained slot space**, which predicts structured gradients rather than unconstrained repair. This is in the spirit of constructional approaches that treat meaning as tied to form–function pairings (Goldberg 1995), while keeping the diagnostic discipline that prevents “anything goes.”

2. What persists across shifts.

A pure contextualist interpretation of type-shifting risks collapsing lexical identity into momentary activation. The exeme→lexeme→lemma layering is introduced to block that collapse: it separates (i) a persistent form-unit (exeme), (ii) a language-specific lexical object (lexeme), and (iii) an indexing convention (lemma), which makes it possible to talk about persistence while still permitting context-sensitive profiling. This is a targeted response to the well-known tension between stable lexical organization and context-dependent interpretation (Lakoff & Johnson 1980).

So, the framework does not deny coercion; it constrains it: coercion becomes a structured set of hosting outcomes, testable by witness regimes and predictable in cost.

8.5 Objection family IV: “ULD is just a metaphor—‘geometry’ is not a theory”

Objection.

Calling something a “domain” or a “geometry” risks being metaphorical. Unless ULD is formally specified and yields unique predictions, it is not explanatory.

Reply.

The objection states the correct standard: geometry-talk is valuable only if it becomes a constraint system. The ULD proposal is introduced precisely to do that work: to treat linguistic activity as occurring within a structured space of admissible positions, with internal organization across levels and with explicit placement constraints that generate predictions about what is licensed where. The point is not rhetorical; it is methodological. It aligns with the general trend in morphosyntax and construction grammar toward deriving category outcomes from structural context (Halle & Marantz 1993; Goldberg 1995), while insisting that “context” must be formalized as an admissible-position space rather than left as an unanalyzed remainder.

ULD becomes non-metaphorical when it produces two kinds of predictions:

- **Placement predictions:** shifting the host position should shift interpretive status in predictable ways (not merely “context affects meaning”).
- **Constraint predictions:** some transitions are cheap and productive, some require overt packaging, and some are effectively blocked—differences that should correlate with acceptability, frequency, processing measures, and computational error patterns.

This is also where expectation-based parsing work becomes relevant: if hosting is constrained, violations should carry measurable processing cost in the same broad family as surprisal-based difficulty effects (Hale 2001; Levy 2008), but with the stronger claim that the cost is tied to *position-forcing* in a slot geometry, not only to linear sequence probability.

8.6 Objection family V: “Cross-linguistic variation undermines universality”

Objection.

Languages differ radically. Some languages are said to have flexible lexical categories; others are not. Some languages lack obvious nominalizers; others require them. Your universality claim risks being Anglocentric or dependent on a small sample.

Reply.

The universality claim is not that every language uses the same markers. It is that languages share a capacity for **objectification as a discourse operation**, and they differ in the surface strategies and **cost profiles** by which that operation is implemented. This is compatible with the general view that category behavior can emerge from structural conditions and constructions rather than from intrinsic lexical labels (Halle & Marantz 1993; Goldberg 1995). It also fits the well-known fact that possession/genitive constructions provide a cross-linguistically robust site for probing noun-like behavior, even though the morphosyntax differs (Partee 1997).

In other words: typological variation becomes a matter of **edge weighting** inside a shared hosting space—what is cheap in one language may be costly in another—rather than a refutation of the underlying slot capacity.

8.7 Objection family VI: “The exeme concept is unnecessary—tokens and lemmas are enough”

Objection.

Why introduce exemes? Linguists and computational systems already use tokens and lemmas. Adding a new unit seems like theoretical inflation.

Reply.

Tokens and lemmas do not solve the identity problem the framework needs to solve.

- A **token** is an event (an occurrence in speech or writing). Tokens are too local to define persistence.
- A **lemma** is an indexing convention (a citation key for dictionaries/corpora). Lemmas are useful but do not explain why a range of realizations count as “the same expression.”
- A **lexeme** is a language-specific lexical unit, but lexemes presuppose that we already know what the stable form-pattern is across modalities and realizations.

An **exeme** is introduced to stabilize the bottom layer: a modality-neutral expression pattern that can be realized in speech, writing, and other media, and that can underwrite lexeme formation and lemma indexing. This is also what makes the framework compatible with both traditional lexicalist concerns (Chomsky 1970) and more dynamic category theories (Halle & Marantz 1993; Goldberg 1995): it provides a persistent unit without forcing intrinsic category membership.

8.8 Objection family VII: “Neuroscience and computation are overclaimed”

Objection.

The book risks overstating neurolinguistic and computational support. Neural data are noisy, and NLP improvements can be achieved by ad hoc heuristics. The theory might be using external domains rhetorically.

Reply.

The correct standard is triangulation: neuroscience and computation are not “proof,” but **constraints and stress tests**. This is consistent with contemporary neurocognitive models that distinguish stored lexical knowledge from combinatorial integration/unification processes (Hagoort et al. 2004), and with distributed semantic accounts linking lexical representations to broader sensorimotor and conceptual systems (Pulvermüller 2005). On this view, the book’s commitments translate into testable expectations:

- If RED captures real stabilization behavior, it should improve parsing/SRL where objectification matters (not everywhere).
- If ULD captures real hosting constraints, violations should correlate with processing cost and error patterns, in the broad family of expectation-based effects (Hale 2001; Levy 2008).
- If the framework is wrong, it should fail in informative ways—e.g., no measurable gradients, no task-specific regime shifts, no computational gain where objectification is the bottleneck.

So, the external chapters are not rhetorical ornaments; they are commitments to accountability.

8.9 What would count as genuine refutation?

A strong theory states not only what it explains but also what would force it to change. On the present framework, the most serious failures would be:

1. **Stable RED-blocking:** classes of items that cannot be stabilized as discourse objects under any admissible witness regime, even in explicit mention contexts, across speakers and registers.

2. **No gradient structure:** acceptability, frequency, and processing measures do not correlate with predicted hosting-cost patterns (e.g., no difference between low-cost and high-cost stabilizations).
3. **ULD non-predictiveness:** explicit slot geometry adds no explanatory or computational value—i.e., it cannot constrain overgeneration or predict where stabilizations are licensed, beyond what existing formalisms already do (Halle & Marantz 1993; Goldberg 1995).
4. **No plausible mapping to known cognitive decompositions:** if the persistence/activation separation yields no alignment with independently motivated distinctions between lexical access and integration/unification (Hagoort et al. 2004; Pulvermüller 2005).

These criteria are intentionally demanding; they define an agenda for testing and revision rather than a rhetorical posture.

8.10 Revisions prompted by critique

Several objections above motivate concrete refinements already built into the book’s architecture:

- The “triviality” objection motivates the **cost/gradient** framing and the separation of ordinary discourse objectification from metalinguistic mention (Partee 1997).
- The “metaphor” objection motivates keeping ULD tied to **explicit constraints** and to measurable predictions in both human processing and computational behavior (Hale 2001; Levy 2008).
- The “identity inflation” objection motivates sharpening the **exeme→lexeme→lemma** triad to prevent persistence claims from collapsing into annotation habits.
- The “overclaiming” objection motivates restricting neuroscience and computation to **triangulation**—externally testable predictions and performance claims, not rhetorical validation (Hagoort et al. 2004; Pulvermüller 2005).

The outcome is a framework that remains ambitious but is not insulated from evidence. It is designed to be wrong in informative ways.

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Chapter 9. Future Directions: From Diagnostic to Geometry

The preceding chapters have advanced two linked claims. First, **RED** is not merely a catalog of nominalization tricks, but a disciplined way of testing when linguistic material can be stabilized as an entity in noun-slot positions. Second, those recurring RED “road signals” are best understood against a stronger background commitment: a **Universal Linguistic Domain (ULD)**—a constrained space of admissible discourse-positions—within which categorization and stabilization occur. In this sense, RED is an evidential procedure *within* a geometry, rather than a free-standing diagnostic applied to a category system from the outside.

What remains is to turn that architecture into a research program: to specify what must be formalized, what must be tested, what can fail, and what would count as genuine confirmation (or disconfirmation) across theory, typology, neuroscience, and computation.

9.1 Theoretical refinements: what ULD still owes the reader

A central theoretical task is to **make ULD explicit as a multi-level generative geometry** rather than a metaphor, including its entry conditions and its internal distance relations. The book’s working picture treats linguistic activity as occurring inside one universal slot-space with internal organization, where items can be “closer” or “farther” from canonical nounal stabilization depending on what operations are required to host them as entities.

(i) A metric for RED-cost.

A promising formal direction is to treat RED as measuring a kind of *distance-to-hosting*: how many (and what kinds of) operations are required to bring a target into noun-slot stabilization. In the ULD discussion, this is already sketched as a distance-to-Ordinate notion (a “RED-cost”) with the expectation that high-cost cases correlate with processing pressure and greater reliance on explicit packaging/mention contexts.

(ii) Reachability vs. conversion.

Future work should sharpen the distinction between (a) *reachability*—an item can be hosted as an entity under the right construction—and (b) *conversion*—a language has conventionalized lexical resources that make such hosting low-cost and default-like. The ULD framing already pushes in this direction by restating RED as detecting when items are “forced” into the argumental/nominal region by constructional pressures rather than by lexical category alone.

(iii) The exeme–lexeme–lemma interface as part of the geometry.

Chapter 3’s layered architecture (expression patterns vs. lexical organization vs. indexing/annotation practice) is not bookkeeping; it is required if ULD is to be computationally implementable and empirically testable. In particular, the **exeme** layer is designed as the modality-neutral, lowest formal unit on which higher lexical and

indexing objects can be defined, while lexemes and lemmas introduce language-internal organization and data-facing conventions, respectively.

9.2 Empirical expansion: typology, corpora, diachrony, modality

The strongest next step is a systematic empirical program that treats RED not as a hand-built demonstration but as a **measurable distribution** across languages, genres, and modalities.

Cross-linguistic typology. The immediate aim is to build a typology of **packaging devices** (possessives, nominalizers, complementizers, classifier-like strategies, prosodic mention cues) and to ask whether ULD predicts stable cross-family regularities in which devices minimize RED-cost for particular domains (events, properties, relations, function-words).

Corpus-scale RED profiling. A corpus program can operationalize RED by annotating candidate constructions and calculating how often non-nominal categories appear in argument positions under identifiable packaging operations (cf. related concerns in argument selection and constructional slotting; Dowty 1991; Goldberg 1995). A key deliverable would be a multi-genre “RED atlas” showing which constructions act as reliable gateways into noun-slot stabilization.

Diachrony and conventionalization. If ULD is a space of possible hosting, historical change should be describable as **pathways through that space**—how certain high-cost stabilizations become grammaticalized (or lexicalized) into low-cost defaults. This connects naturally to category-neutral root approaches (Halle & Marantz 1993) but differs by making “slot-space” explicit and measurable.

Modality and media. Because the exeme layer is explicitly modality-neutral, the program should include tests across speech, writing, and digitally mediated text—asking whether “entity stabilization” behaves similarly when the packaging cue is morphological, orthographic, prosodic, or layout-based. This is also where exeme inventories and alignment become crucial as a shared substrate for cross-modal comparison.

9.3 Neuroscience and psycholinguistics: predicted signatures of slot-stabilization

A central promise of the book is that ULD is not merely descriptive; it has **processing and neural consequences**. If RED-cost is meaningful, then increasing distance-to-hosting should show up as increased unification/repair and prediction difficulty during comprehension.

One concrete prediction already stated in the ULD discussion is that **high RED-cost** cases should correlate with stronger ERP signatures often associated with syntactic reanalysis/unification pressure (e.g., P600-type effects) and with stronger sensitivity to packaging/mention contexts.

This can be developed into a testable experimental package:

- **ERP/MEG comprehension studies** contrasting low-cost nominalizations (conventional gerunds, standard possessives) with deliberately high-cost cases (nominalized function-words, marginal possessives) to test graded effects (Kutas & Hillyard 1980; Bemis & Pykkänen 2011).
- **fMRI unification/predication contrasts** targeting whether noun-slot stabilization recruits predictable unification networks (e.g., work associated with unification models; Hagoort 2013) and whether the gradient aligns with a RED-cost metric rather than with surface part-of-speech labels.
- **Cross-linguistic processing comparisons** testing whether languages with different packaging inventories produce different cost profiles for comparable “entity stabilization” tasks, consistent with the idea that languages implement different low-cost routes through the same universal slot-space.

9.4 Computational agenda: ULD-aware RED as an engineering target

The computational program is not an optional application; it is a way of forcing precision. A diagnostic that cannot be implemented is difficult to falsify. The ULD framing suggests a clear engineering direction: **treat**

RED as a detect-and-update operation over a slot-structured representation rather than as a post-hoc interpretation of surface strings.

Key deliverables:

(i) Benchmark tasks explicitly tied to noun-slot stabilization.

Beyond general parsing accuracy, build evaluation sets where success requires recognizing that a non-canonical category is functioning as an entity (e.g., clausal subjects, packaged predicates, nominalized connectives). This directly targets the gap between category labels and slot-function.

(ii) RED-enhanced parsing and SRL.

Integrate RED detection into dependency parsing and semantic role labeling pipelines (Palmer et al. 2005), using ULD-style slot updates as an intermediate representation. The hypothesis is that explicitly modeling “entity stabilization” improves robustness on constructions that standard tag-and-parse pipelines mis-handle.

(iii) Probabilistic processing models as a bridge to RED-cost.

If RED-cost corresponds to processing difficulty, then surprisal-based models (Hale 2001; Levy 2008) provide a principled way to relate ULD distance metrics to distributional expectations, and neural encoders can be probed for whether they implicitly learn “slot stabilization” behavior (Devlin et al. 2019; Conneau et al. 2020).

9.5 What would count as failure: falsifiability conditions

A future-directions chapter should also state what could prove the framework wrong. At minimum, the following would force revision:

1. **No stable gradient:** if carefully controlled studies fail to find any systematic relation between RED-cost (operationalized) and behavioral/neural measures, the “distance-to-hosting” idea would be weakened.
2. **Strong counter-typology:** if broad typological work identifies languages where entity stabilization does *not* correlate with identifiable packaging routes, or where noun-slot behavior is sharply restricted in ways ULD cannot model without ad hoc stipulation.
3. **No computational gain under controlled benchmarks:** if RED-aware systems do not improve on stabilization-sensitive constructions (or cannot be made precise enough to implement without collapsing into unconstrained heuristics), then the diagnostic/geometry link needs rethinking.

9.6 Closing: from universal noun slot to a general theory of discourse-space

The long-range goal is to generalize the core insight: the universal noun slot is not a curiosity about nominalization, but evidence that linguistic cognition is organized around a **space of possible hosting positions**—a geometry in which materials are stabilized as entities, predicates, modifiers, and discourse operators under constructional and contextual pressure. RED is the entry method into that geometry; ULD is the attempt to state the geometry itself.

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Chapter 10

Conclusion: From Category Labels to Hosting Geometry

10.1 What this book set out to explain

This book began from a familiar tension in linguistic theory. On the one hand, grammatical categories are often treated as stable lexical types—nouns as natural arguments, verbs as natural predicates, adjectives as natural property terms (Chomsky 1970). On the other hand, ordinary linguistic practice repeatedly and systematically violates that division: speakers turn actions into “things,” properties into “objects of thought,” relations into entities, and even grammatical function-words into discussable items (Lakoff & Johnson 1980). The descriptive facts are not difficult to assemble; the difficulty is explaining how such flexibility can be *reliable* and *structured* without reducing the category system to triviality.

The book’s central claim is that reliability is not accidental. It reflects a constrained architecture in which linguistic material can be stabilized in particular **discourse positions**—positions that behave like slots for entities, predicates, modifiers, operators, and linkers. The core contribution is therefore not simply to reiterate that “categories are flexible,” but to propose a way to make that flexibility **diagnosable** and **geometrically constrained**.

10.2 What RED contributes: a disciplined evidential method

The first contribution is methodological. The **Reverse Entailment Diagnostic (RED)** was introduced as a bidirectional inference tool designed to test **nominal potential**: whether a linguistic item can be stabilized as a syntactic noun entity that occupies noun-slot positions (subject/object/NP-internal) under identifiable witness regimes (e.g., possessives/genitives, nominalizers, controlled mention/quotation contexts) (Dowty 1991; Partee 1997). This diagnostic is motivated by the broader shift away from intrinsic category essentialism and toward approaches in which category outcomes emerge from structural contexts and functional heads (Halle & Marantz 1993; Goldberg 1995).

RED's key value is discipline. It prevents the universal noun-slot claim from collapsing into vague slogans by requiring concrete evidence: a packaging witness, an interpretability/grammaticality check, and a verification step in noun-slot occupancy. In doing so, RED turns “category flexibility” into a phenomenon with testable boundaries and gradable outcomes rather than an open-ended invitation to reinterpretation.

10.3 Why RED forces ULD: the need for a hosting geometry

As the argument developed, it became clear that a diagnostic cannot float in isolation. RED presupposes a theory of what kind of object is being diagnosed, what counts as evidence, and—most importantly—what kind of structural space makes “stabilization” possible in the first place. Put differently: the repeated RED “road signals” (nominal potential → noun-slot occupancy → entity stabilization) point beyond the diagnostic to a background architecture.

That background architecture is what the book calls the **Universal Linguistic Domain (ULD)** (or ULD/UNS): a constrained space of admissible syntactic–cognitive positions within which linguistic material can be hosted and stabilized. The universal noun slot is thus not a loose description of flexibility, but a **domain-level constraint**: a privileged region of the hosting space where diverse material can be stabilized as entity-like when constructions demand it.

In this architecture, **RED is the evidential probe**, and **ULD is the geometry** the probe is probing. This pairing is the book's main conceptual advance: it replaces an opposition between “rigid categories” and “free contextual fluidity” with a third option—**structured placement** in a constrained domain.

10.4 The lexical layering: exeme, lexeme, lemma

A further contribution of the book is to separate what linguistics often conflates: the persistent unit of expression, the language-specific lexical organization, and the indexical conventions by which dictionaries and corpora name lexical entries. This is the motivation for the **exeme–lexeme–lemma** triad.

- The **exeme** provides the lowest formal layer: a modality-neutral expression pattern that can be realized in speech, writing, and other media, and that can underwrite lexical identity across variation.
- The **lexeme** is the language-specific projection of exemic material into morphosyntactic systems—paradigms, argument structure, conventionalized category profiles.
- The **lemma** is the citation/indexing key that lexicography and annotation impose for reference and retrieval.

This layering does not replace existing practice; it clarifies what each practice is actually tracking. It also supplies a necessary interface between theory and implementation: computational systems need an explicit account of what is persistent and what is context-activated (Devlin et al. 2019), and ULD provides a principled location to express that separation.

10.5 Convergence beyond theory: cognition, brain, and computation

The book has also argued that the RED→ULD transition is not merely a formal maneuver. It carries implications that can be tested in cognitive science and computation.

On the cognitive side, the universal noun slot hypothesis aligns with a broad view in cognitive linguistics that conceptualization routinely objectifies events and properties as “things” for purposes of discourse management (Lakoff & Johnson 1980). Neurocognitively, the book’s prediction is that “entity stabilization” has measurable signatures: cases that force non-canonical material into noun-slot positions should carry integration costs and recruit combinatory/unification operations beyond default lexical access, in line with unification-oriented models and distributed lexical-semantic activation (Hagoort et al. 2004; Pulvermüller 2005).

On the computational side, the book has argued that RED is implementable as a detect-and-update mechanism in parsing and semantic role labeling pipelines: identify packaging witnesses, promote packaged spans into argumental nodes, and then compute role structure over the updated representation (Palmer et al. 2005).

The broader point is methodological: computational implementation forces the theory to be explicit about what counts as evidence and what constraints are operative, and it supplies an independent evaluation channel for whether ULD-like constraints add predictive value beyond surface POS tagging.

10.6 What this changes in the landscape of approaches

The book’s position relative to existing work can be summarized as follows.

1. **Against intrinsic category essentialism:** categories are not treated as stable lexical types assigned once and for all (Chomsky 1970).
2. **Beyond unconstrained contextualism:** context does not license arbitrary reinterpretation; it selects among structurally admissible placements within a constrained domain.
3. **Compatible with dynamic morphosyntax and constructions, but more explicit about hosting:** root-neutral and constructional approaches are correct to treat categories as outcomes of structure (Halle & Marantz 1993; Goldberg 1995), but the present framework attempts to make the space of outcomes explicit as a geometry of positions and transitions.
4. **Operational and testable:** RED turns category potential into a measurable diagnostic; ULD turns the diagnostic’s success into evidence about domain-level constraints.

10.7 Final statement

The universal noun slot claim, as argued here, is not the claim that “everything is a noun.” It is the claim that linguistic cognition and grammar make available a constrained region of discourse space in which diverse material can be stabilized as an entity, and that this stabilization is visible through disciplined diagnostics. RED shows that stabilization occurs; ULD explains how it is possible and why it is constrained.

If the book’s architecture is correct, then future work will not merely multiply examples. It will build maps: cross-linguistic maps of packaging regimes, quantitative maps of stabilization cost, neurocognitive maps of hosting pressures, and computational systems that treat placement—not intrinsic category labels—as the fundamental explanatory unit. In that sense, the book’s conclusion is also its invitation: to treat linguistic categories as *products of hosting geometry*, and to test that geometry with the same rigor we apply to any other claim about the structure of the linguistic mind.

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Chapter 11 — References and Appendices

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Appendix A — Formal Architecture of the Universal Linguistic Domain

A.1 Objects and levels of organization

This book assumes a stratified architecture in which increasingly complex linguistic organizations are built from simpler entry units:

Level-0 (Entry): Exeme tokens.

An **exeme** is the modality-neutral expression unit: a form-pattern (e.g., a segmental/graphic sequence) abstracted away from grammatical category assignment. At Level-0, an exeme is treated as an input object eligible to be hosted.

Level-1 (Primary organization): Term (three-region tape).

A **term** is the minimal hosting configuration that stabilizes an exeme as an interpretable discourse object inside a three-region structure:

- **Ordinate** (nounal nucleus; the entity-stabilizing core)
- **Subordinate** (determiner/quantificational/qualitative sequencing)
- **Coordinate** (framing/relational material hosted term-internally)

Notation: a term may be represented schematically as $t = \langle \text{Coord}, \text{Subord}, \text{Ord} \rangle$, with unfilled positions marked $\emptyset$.

Level-2 (Secondary organization): Two-term predication and PAS.

Two Level-1 terms can be linked by a predication relation **R** into a **two-term structure** $t_1 R t_2$. In its canonical expansion, this relation yields the **predicate–argument skeleton (PAS)**: the minimal generative frame from which “ready-to-go” syntactic units are derived.

At this level, the book treats **phrases, clauses, sentences, and certain compound words** as **derived realizations** of PAS: they are not the primitive starting point, but structured elaborations of the same underlying predication skeleton.

A.2 Notational conventions for term annotation

Examples in Chapters 3–6 may be annotated with bracketed region specifications, distinguishing Coordinate/Subordinate/Ordinate fillings, and marking empty positions with $\emptyset$. Multiple exeme tokens co-hosted in a micro-position may be written as comma-separated sequences. (See also Appendix C for full annotated example lists.)

A.3 The position function and positional category

Let ϕ be a **position function** mapping hosted material to micro-positions in the term/connector space. Category is treated as **positional** rather than intrinsic: an exeme’s realized grammatical profile is the result of **where** it is hosted (Ordinate vs Subordinate vs Coordinate; and, at Level-2, which connector/axis micro-slots it occupies).

A.4 RED as graded Ordinate reachability

RED is formalized as a function that takes a base item (or construction) **X** and a set of nominalizing environments/derivations **Y**, returning a **graded output** reflecting:

1. **Ordinate reachability** (can X be stabilized as an entity so that noun-slot occupancy is licensed?), and
2. **hosting stability / markedness cost** (how “expensive” or metalinguistically constrained the stabilization is).

This appendix defines the output scale used throughout the book (e.g., strong positive; weak-but-positive; marginal; negative), and the evidential conditions required for each label.

Appendix B — RED Test Suite and Controls

This appendix provides the standardized RED battery used across chapters:

1. **Derivational probes** (possessive markers, nominalizers, determiner environments).
2. **Reverse entailment step** ($Y \rightarrow X$), with explicit acceptance criteria.
3. **Noun-slot verification** (subject, object, predicate nominal, topic positions).
4. **Controls for “use vs. mention”** (metalinguistic regimes, quotation/orthographic mention, classroom-definition contexts).
5. **Function-word edge cases** (articles, conjunctions, prepositions, interjections), with explicit criteria for when the data count as grammatical evidence versus pure mention.

Templates are provided in a language-neutral format, with language-specific instantiations for English, Mandarin, Tongan, Spanish, and Arabic.

Appendix C — Data Inventory, Example IDs, and Annotation Schema

This appendix consolidates the book’s empirical materials into a reusable dataset:

- **Example IDs** (Chapter.Section.ExampleNumber)
- **Language code** and dialect/variety notes
- **Source type** (constructed; corpus-attested; elicited judgment)
- **Gloss line(s)** and translation line
- **RED outcome label** (graded) and brief justification
- **ULD/term annotation** (region fillings; Ø markings; connector placement where relevant)

Where judgments are used, this appendix specifies the elicitation conditions and the interpretation standard (e.g., ordinary discourse vs explicitly metalinguistic contexts).

Appendix D — Algorithmic Specification (RED in NLP Pipelines)

This appendix gives an implementation-ready specification for integrating RED into parsing and role labeling:

1. **Nominalization candidate detection** (marker inventory per language; phrase-level heuristics).
2. **Reverse-entailment confirmation** (base extraction; grammaticality filters; metalinguistic gating).
3. **Dependency/constituency updates** (re-typing nominalized constituents as candidate noun-slot occupiers).
4. **PAS-aware SRL adjustments** (role assignment improvements in the presence of nominalized events/properties).

5. **Evaluation protocol** (precision/recall/F1; multilingual transfer testing; edge-case reporting).

Appendix E — Neurolinguistic Prediction Set and Minimal Experimental Designs

This appendix states the book’s neuroscience-facing claims in testable form:

- **Prediction class 1 (Convergence):** nominal environments recruit overlap between entity-processing and action/property networks when non-nominal bases are stabilized as noun-slot occupiers.
- **Prediction class 2 (Unification cost):** graded RED outcomes correlate with increased unification demands (e.g., IFG engagement) as nominal stabilization becomes more marked.
- **Minimal designs:** sentence-level paradigms contrasting (i) ordinary nominal uses, (ii) nominalized event/property uses, (iii) metalinguistic mention controls, suitable for EEG/MEG or fMRI.

Appendix F — Abbreviations and Glossing Conventions

This appendix lists:

- **Book-wide abbreviations** (ULD, UNS, RED, PAS, SRL, etc.)
- **Glossing abbreviations** (e.g., GEN, POSS, DET, CLF, NOM, etc.)
- **Example formatting standard** (numbering, alignment, translation conventions)